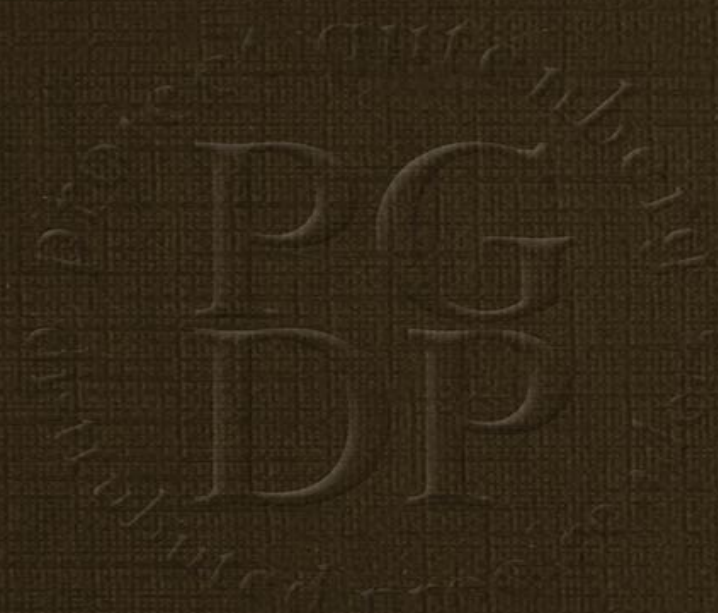


THE
CELESTIAL
WORLDS
DISCOVER'D



CHRISTIAAN HUYGENS
1722

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Title: The celestial worlds discover'd
or, conjectures concerning the inhabitants, plants and productions of the
worlds in the planets

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Editor: Constantijn Huygens

Translator: John Clarke

Release date: July 14, 2023 [eBook #71191]

Language: English

Original publication: United Kingdom: James Knapton, 1722

Other information and formats: www.gutenberg.org/ebooks/71191

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*** START OF THE PROJECT GUTENBERG EBOOK THE CELESTIAL WORLDS
DISCOVER'D ***

THE
Celestial Worlds
DISCOVER'D:
OR,
CONJECTURES
Concerning the
INHABITANTS,
PLANTS and PRODUCTIONS
OF THE
Worlds in the Planets.

Written in Latin by
CHRISTIANUS HUYGENS,
And inscrib'd to his Brother
CONSTANTINE HUYGENS
Late Secretary to his Majesty King *William*.

The Second Edition, Corrected and Enlarged.

L O N D O N:
Printed for JAMES KNAPTON, at the *Crown* in
St. Paul's Church-Yard. MDCCLXXII.

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TO THE READER.

This Book was just finished, and designed for the Press, when the Author, to the great loss of the Learned World, was seized by a Disease that brought him to his Death. However he took care in his last Will of its Publication, desiring his Brother, to whom it was writ, to take that Trouble upon him. But he was so taken up with Business and Removals, (as being Secretary in *Holland* to the King of *Great Britain*) that he could find no time for it till a Year after the Death of the Author: When it so fell out, that the Printers being somewhat tardy, and this Gentleman dying, the Book was left without either Father or Guardian. Yet it [iv] now ventures into the Publick, in the same Method that it was writ by the Author, and with the same Inscription to his Brother, tho' dead; in confidence that this last Piece of his will meet with as kind a Reception from the World as all the other Works of that Author have. 'Tis true there are not every where Mathematical Demonstrations; but where they are wanting, you have probable and ingenious Conjectures, which is the most that can be reasonably expected in such matters. What belongs to, or has any thing to do with Astronomy, you will see demonstrated, and the rest ingeniously and shrewdly guessed at, from the Affinity and Relation of the heavenly Bodies to the Earth. For your farther Satisfaction read on, and farewell. [v]

THE
PUBLISHER
TO THE
READER.

I Doubt not but I ſhall incur the Cenſures of learned Men for putting this Book into Engliſh, becauſe, they'll ſay, it renders Philoſophy cheap and vulgar, and, which is worſe, furniſhes a ſort of injudicious People with a ſmattering of Notions, which being not able to make a proper uſe of, they pervert to the Injury of Religion and Science. I confeſs the Allegation is too true: but after Biſhop Wilkins, Dr. Burnet, Mr. Whiſton and others, to ſay nothing of the ancient Philoſophers, who wrote in their own [vi] Tongues; I ſay, after theſe great Authors have treated on as learned and abſtruſe Subjects in the ſame Language, I hope their Example will be allowed a ſufficient excuſe for printing this Book in Engliſh.

Concerning this Edition I can ſay, that I have taken care to have the Cutts exactly done, and have placed each Figure at the Page of the Book that refers to it, which I take to be more convenient to the Reader than putting them all at the End.

I have been careful to procure the beſt Paper, that I might in ſome meaſure come up to the Beauty of the Latin Edition, though this bear but half the Price of it.

And I hope the Tranſlator has expreſſed the Author's Senſe aright, and has not committed Faults beyond what an ingenuous Reader can pardon. [1]

NEW
CONJECTURES
Concerning the
Planetary Worlds,
THEIR
INHABITANTS
AND
PRODUCTIONS.

*Written by CHRISTIANUS HUYGENS,
and inscribed to his Brother
CONSTANTINE HUYGENS.*

BOOK the Firft.

A Man that is of *Copernicus's* Opinion, that this Earth of ours is a Planet, carry'd round and enlighten'd by the Sun, like the reft of the Planets, cannot but sometimes think, that it's [2]not improbable that the reft of the Planets have their Drefs and Furniture, and perhaps their Inhabitants too as well as this Earth of ours: Efpecially if he confiders the later Difcoveries made in the Heavens fince *Copernicus's* time, viz. the Attendants of *Jupiter* and *Saturn*, and the champaign and hilly Countries in the Moon, which are a ftrong Argument of a Relation and Kin between our Earth and them, as well as a Proof of the Truth of that Syftem. This has often been our Talk, I remember, good Brother, over a large Telescope, when we have been viewing thofe Bodies, a Study that your continual Bufinels and Abfence have interrupted for many Years. But we were always apt to conclude, that 'twas in vain to enquire after what Nature is doing there, feeing there was no likelihood of ever coming to any Certainty of the Enquiry. Nor could I ever find that any Philofophers, either antient or modern, have attempted any thing upon this Subject. At the very Birth [3]of Aftronomy, when the Earth was firft afferted to be Spherical, and to *Some have already talk'd of the Inhabitants of the Planets, but went no farther.* be furrounded with Air, even then there were fome Men fo bold as to affirm, there were an innumerable Company of Worlds in the Stars. But later Authors, fuch as Cardinal *Cufanus*, *Brunus*, *Kepler*, (and if we may believe him, *Tycho* was of that opinion too) have furnifhed the Planets with Inhabitants. Nay, *Cufanus* and *Brunus* have allowed the Sun and fixed Stars theirs too. But this was the utmoft of their Boldnefs; nor has the ingenious *French* Author of the Dialogues about *the Plurality of Worlds* carried this Matter any farther. Only fome of them have coined fome Stories of the Men in the Moon, juft as probable as *Lucian's* true Hiftory; among which I muft count *Kepler's*, which he has diverted us with in his Aftronomical Dream. But a while ago thinking fomewhat ferioufly of this matter (not that I count my felf quicker-fighted than thofe great Men, but that I had the Happinefs to live after [4]moft of them) the Enquiry appeared not fo impracticable, nor the Way fo ftopt up with Difficulties, but that there was very good room left for probable Conjectures. As they came into my Head, I put them down into common Places, and fhall now try to digeft them into fome Method for your better Conception of them, and add fomewhat of the Sun and fix'd Stars, and the Extent of that Univerfe of which our Earth is but an inconfiderable Point. I know you have fuch an Esteem and Reverence for any thing that belongs to the Heavens, that I perfwade my felf you will read what I have written with fome Pleafure: I'm fure I writ it with a great deal; but as often before, fo now, I find the Saying of *Archytas* true, even to the Letter, *That tho' a Man were*

admitted into Heaven to view the wonderful Fabrick of the World, and the Beauty of the Stars, yet what would otherwise be Rapture and Extasie, would be but a melancholy Amazement if he had not a Friend to communicate [5]it to. I could wish indeed that all the World might not be my Judges, but that I might chuse my Readers, Men like you, not ignorant in Astronomy and true Philosophy; for with such I might promise my self a favourable hearing, and not need to make an Apology for daring to vent any thing new to the World. But because I am aware what weak Hands it's likely to fall into, and what a severe Sentence I may expect from those whose Ignorance or Zeal is too great; it may be worth the while to guard my self beforehand against the Assaults of those sort of People.

The Objections of ignorant Cavillers prevented. There's one sort who knowing nothing of Geometry or Mathematicks, will laugh at it as a whimsical and ridiculous Undertaking. It's an incredible Thing to them to talk of measuring the Distance and Magnitude of the Stars: And for the Motion of the Earth, they count it, if not a false, at least a precarious Opinion; and no wonder then if they take what's built upon such a slippery Foundation [6]for the Dreams of a fanciful Head and a distemper'd Brain. What should we answer to these Men, but that their Ignorance is the Cause of their Dislike, and that if they had studied these things more, and viewed the Works of Nature nicely, they would have fewer Scruples? But few People having had an opportunity of prosecuting these Studies, either for want of Parts, Learning or Leisure, we cannot blame their Ignorance; and if they resolve to find fault with us for spending time in such Matters, because they do not understand the Use of them, we must appeal to proper Judges.

These Conjectures do not contradict the holy Scriptures. The other sort, When they hear us talk of new Lands, and Animals, and Creatures endued with as much Reason as themselves, will be ready to cry out, that we set up our Conjectures against the Word of God, and broach Opinions directly opposite to Holy Writ. For we do not there read any thing of the Production of such Creatures, no not so much as that they exist; nay rather we [7]read the quite contrary. For, That only mentions this Earth with its Animals and Plants, and Man the Lord of them: To such Persons I answer, what has been often urged by others before me: That it's evident, God had no design to make a particular Enumeration in the Holy Scriptures, of all the Works of his Creation. When therefore it is plain that under the general Name of *Stars* or *Earth* at the Creation, are comprehended all the Heavenly Bodies, even the Attendants upon *Jupiter* and *Saturn*, why must all that Multitude of Beings which the Almighty Creator has been pleased to place upon them, be excluded the Privilege, and not suffered to have a Share in the Expression? And these Men themselves can't but know in what Sense it is that all things are said to be made for the Use of Man, not certainly for us to look at through a Telescope, for that's very absurd. Since then the greatest part of God's Creation, that innumerable multitude of Stars, is

placed out of the [8]reach of any Man's Eye; and many of them it's likely, of the best Glasses, so that they don't seem to belong to us; is it such an unreasonable Opinion to think, that there are some reasonable Creatures who see and admire those glorious Bodies at a nearer distance?

This Enquiry not over curious. But perhaps they'll say, it does not become us to be so curious and inquisitive in these Things which the Supreme Creator seems to have kept for his own Knowledge: For since he has not been pleased to make any farther Discovery or Revelation of them, it seems little better than presumption to make any inquiry into that which he has thought fit to hide. But these Gentlemen must be told, that they take too much upon themselves when they pretend to appoint how far and no farther Men shall go in their Searches, and to set bounds to other Mens Industry; as if they knew the Marks that God has placed to Knowledge: or as if Men were able to pass those Marks. If our Forefathers had been at this rate scrupulous, we might [9]have been ignorant still of the Magnitude and Figure of the Earth, or that there was such a Place as *America*. We should not have known that the Moon is lightened by the Sun's Rays, nor what the Causes of the Eclipses of each of them are, nor a multitude of other Things brought to light by the late Discoveries in Astronomy. For what can a Man imagine more abstruse, or less likely to be known, than what is now as clear as the Sun? Whence it follows, that vigorous Industry, and piercing Wit were given Men to make Advances in the Search of Nature, and there's no Reason to put any Stop to such Enquiries. I must acknowledge that what I here intend to treat of is not of that Nature as to admit of a certain Knowledge; I can't pretend to assert any thing as positively true (for how is it possible) but only to advance a probable Guess, the Truth of which every one is at his own liberty to examine. If any one therefore shall gravely tell me, that I have spent my Time idly in a vain and fruitless Enquiry [10]after what by my own acknowledgment I can never come to be sure of; The Answer is, that at this rate he would put down all Natural Philosophy as far as it concerns it self in searching into the Nature *Conjectures not useles, because not certain.* of Things: In such noble and sublime Studies as these, 'tis a Glory to arrive at Probability, and the Search it self rewards the Pains. But there are many degrees of Probable, some nearer Truth than others, in the determining of which lies the chief exercise *These Studies useful to Religion.* of our Judgment. But besides the Nobleness and Pleasure of the Studies, may not we be so bold as to say, they are no small help to the Advancement of Wisdom and Morality? so far are they from being of no use at all. For here we may mount from this dull Earth, and viewing it from on high, consider whether Nature has laid out all her Cost and Finery upon this small Speck of Dirt. So, like Travellers into other distant Countries, we shall be better able to judge of what's done at home, know how to make a true Estimate of, and let [11]its own Value upon every Thing. We shall be less apt to admire what this World

calls Great, shall nobly despise those Trifles the generality of Men set their Affections on, when we know that there are a multitude of such Earths inhabited and adorned as well as our own. And we shall worship and reverence that God the Maker of all these things; we shall admire and adore his Providence and wonderful Wisdom which is displayed and manifested all over the Universe, to the Confusion of those who would have the Earth and all things formed by the shuffling Concourse of Atoms, or to be without beginning. But to come to our Purpose.

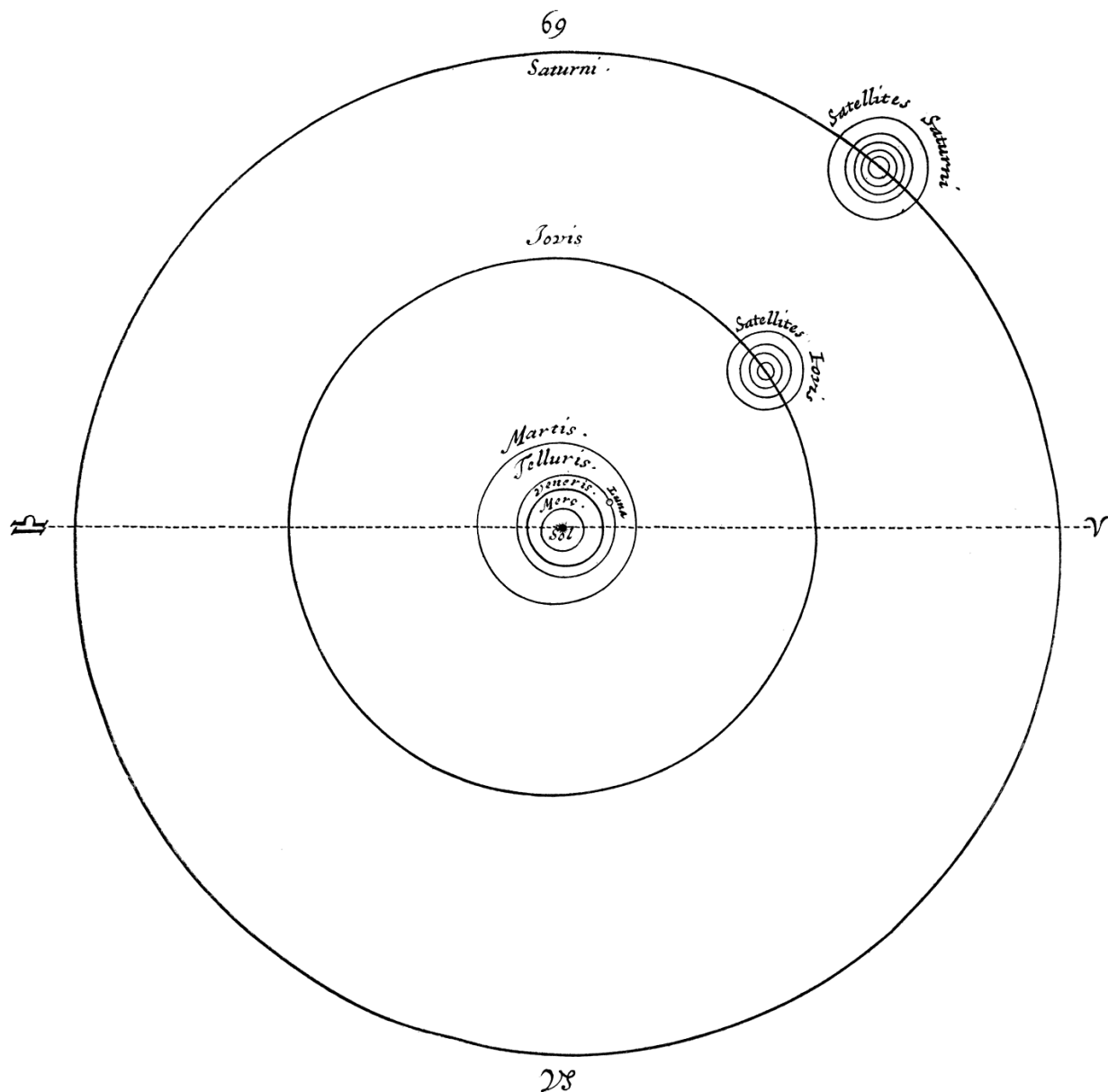


Fig. 1 SYSTEMA COPERNICI.

Copernicus's System explained. And now because the chief Argument for the Proof of what we intend will be taken from the Disposition of the Planets, among which without doubt, the Earth must be counted in the Copernican System, I shall here first of all draw two Figures. The first is a Description of [12]the Orbs the Planets move in, in that order that they are placed round the Sun, drawn as near as can be in their true Proportions, like what you have seen in my Clock at home. The second shows the Proportions of their Magnitudes in respect of one another and of the Sun, which you know is upon that same Clock of mine too. In the first the middle Point or Center is the Place of the Sun, round which, in an order that every one knows, are the Orbits of *Mercury*, *Venus*, the Earth with that of the Moon about it; then those of *Mars*, *Jupiter* and *Saturn*: and about the two last the small Circles that their Attendants move in: about *Jupiter* four, and about *Saturn* five. Which Circles as well as that of the Moon are drawn larger than their true Proportion would admit, otherwise they could not have been seen. You may easily apprehend the Vastness of these Orbits by this, that the distance of the Earth from the Sun is ten or twelve thousand of the Earth's Diameters. Almost all these Circles are in the same Plane, declining very [13]little from that in which the Earth moves, call'd *The Plane of the Ecliptick*. This Plane is cut obliquely by the Axis upon which the Earth turns it self round with respect to the Sun in 24 Hours, whence arise the Successions of Day and Night: The Axis of the Earth always keeping the same Inclination to the Ecliptick (except a small Change best known to Astronomers) while the Earth itself is carried in its yearly Course round the Sun, causes the regular Order of the Seasons of the Year: as you may see in all Astronomers Books. Out of which I shall transcribe hither the Periods of the Revolutions of the Planets, viz. *Saturn* moves round the Sun in 29 Years, 174 Days, and 5 Hours: *Jupiter* finishes his Course in 11 Years, 317 Days, and 15 Hours: *Mars* his in about 687 Days. Our Year is 365 Days 6 Hours: *Venus*'s 224 Days 18 Hours: and *Mercury*'s 88 Days. This is the now commonly received System, invented by *Copernicus*, and very agreeable to that frugal Simplicity Nature shows in all [14]*Arguments for the Truth on't*. her Works. If any one is resolved to find fault with it, let him first be sure he understands it. Let him first see in the Books of Astronomers with how much greater Ease and Plainness all the Motions of the Stars, and Appearances in the Heavens are explained and demonstrated in this than either in that of *Ptolomy* or *Tycho*. Let him consider that Discovery of *Kepler*, that the Distances of the Planets from the Sun, as well of the Earth as the rest, are in a fix'd certain proportion to the Times they spend in their Revolutions. Which Proportion it's since observed that their Satellites keep round *Jupiter* and *Saturn*. Let him examine what a contradictory Motion they are fain to invent for the Solution of the Polar Star's changing its Distance from the Pole. For that Star in the end of the little Bear's Tail which now describes so small a Circle round the Pole, that it is not above two Degrees and twenty Minutes, was observed about 1820 Years ago, in the Time

of *Hipparchus*, to be above 12: [15]and will within a few Ages more be 45 Degrees distant from it: and after 25000 Years more will return to the same Place it is now in. Now if with them we allow the Heavens to be turned upon their own Axis, at this rate they must have a new Axis every Day: a Thing most absurd, and repugnant to the Nature of all Motion. Whereas nothing is easier with *Copernicus* than to give us Satisfaction in this Matter. Then he may impartially weigh those Answers that *Galilæus*, *Gaffendus*, *Kepler*, and others have given to all Objections proposed, which have so satisfied all Scruples, that generally all Astronomers now-a-days are brought over to our Side, and allow the Earth its Motion and Place among the Planets. If he cannot be satisfied with all this, he is either one whose Dulness can't comprehend it, or who has his Belief at another Man's Disposal.

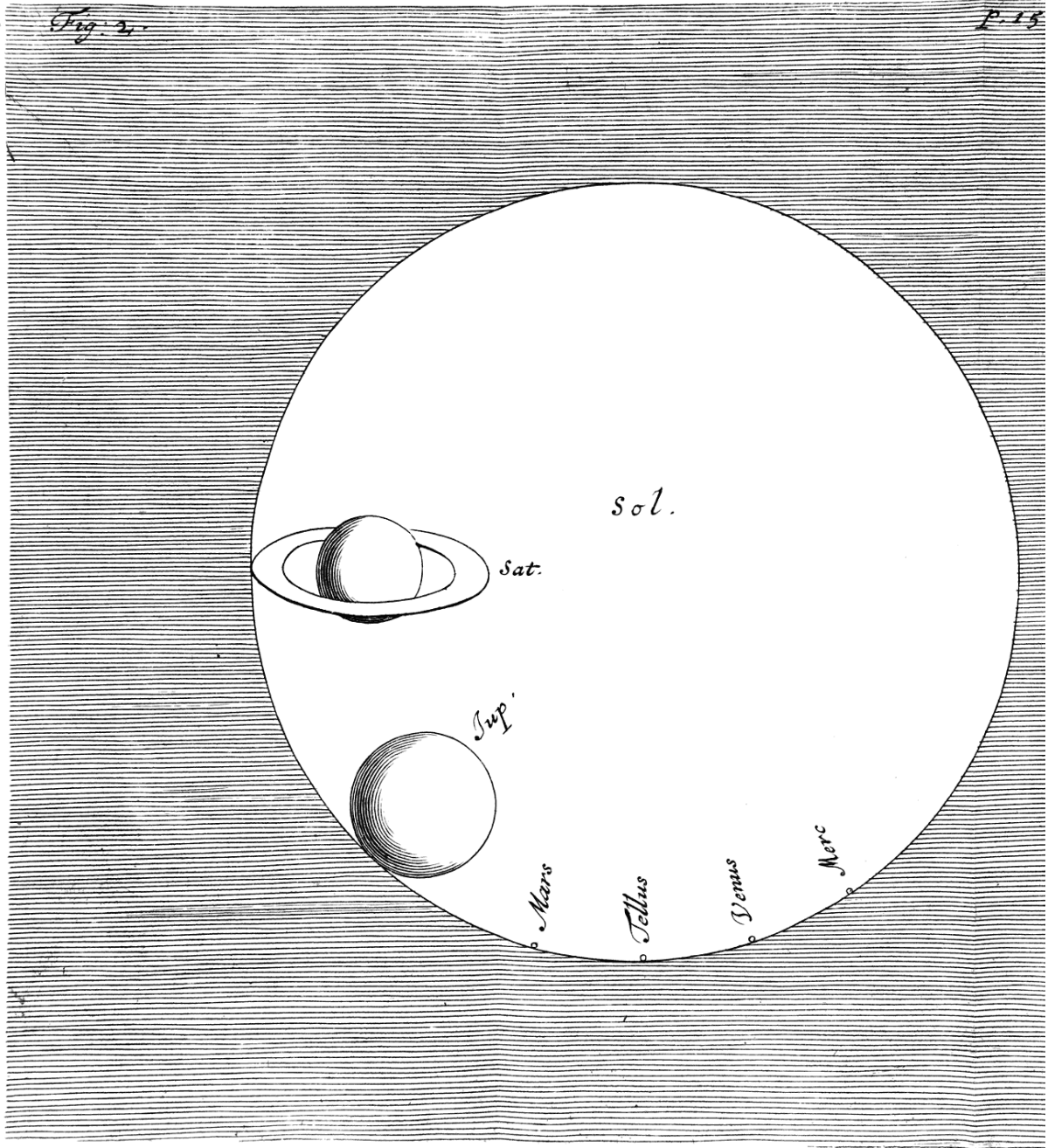


Fig. 2.

p. 15.

In the other Figure you have the Globes of the Planets, and of the Sun, represented to your Eyes as placed near one another. Where [16] *The Proportion of the Magnitude of the Planets, in respect of one another, and the Sun.* I have observed the same Proportion, of their Diameters to that of the Sun, that I published to the World in my Book of *The Appearances of Saturn*: namely, the Diameter of the Ring round *Saturn* is to that of the Sun as 11 as to 37; that of

Saturn himself about as 5 to 37; that of *Jupiter* as 2 to 11; that of *Mars* as 1 to 166; of the Earth as 1 to 111; and of *Venus* as 1 to 84: to which I shall now add that of *Mercury* observed by *Hevelius* in the Year 1661, but calculated by my self, and found to be as 1 to 290.

If you would know the way that we came to this Knowledge of their Magnitudes, by knowing the Proportion of their Distances from the Sun, and the Measures of their Diameters, you may find it in the Book before-mentioned: And I cannot yet see any Reason to make an Alteration in those I then settled, altho' I will not say they are without their Faults. *The Lamellæ more convenient than Micrometers.* For I can't yet be of their Mind, who think the Use of Micrometers, as they call them, is beyond that of [17]our Plates, but must still think that those thin Plates or Rods of which I there taught the Use, not to detract from the due Praises of so useful an Invention, are more convenient than the Micrometers.

In this proportion of the Planets it is worth while to take notice of the prodigious Magnitude of the Sun in comparison with the four innermost, which are far less than *Jupiter* and *Saturn*. And 'tis remarkable, that the Bodies of the Planets do not increase together with their Distances from the Sun, but that *Venus* is much bigger than *Mars*.

The Earth justly likened to the Planets, and the Planets to it. Having thus explained the two Schemes, there's no Body I suppose but sees, that in the first the Earth is made to be of the same sort with the rest of the Planets. For the very Position of the Circles shows it. And that the other Planets are round like it, and like it receive all the Light they have from the Sun, there's no room (since the Discoveries made by Telescopes) to doubt. Another Thing they are like it in is, that they are moved [18]round their own Axis; for since 'tis certain that *Jupiter* and *Saturn* are, who can doubt it of the others? Again, as the Earth has its Moon moving round it, so *Jupiter* and *Saturn* have theirs. Now since in so many Things they thus agree, what can be more probable than that in others they agree too; and that the other Planets are as beautiful and as well stock'd with Inhabitants as the Earth? Or what shadow of Reason can there be why they should not?

If any one should be at the Dissection of a Dog, and be there shewn the Intrails, the Heart, Stomach, Liver, Lungs and Guts, all the Veins, Arteries and Nerves; could such a Man reasonably doubt whether there were the same Contexture and Variety of Parts in a Bullock, Hog, or any other Beast, tho' he had never chanc'd to see the like opening of them? I don't believe he would. Or were we thoroughly satisfy'd in the Nature of one of the Moons round *Jupiter*, should not we straight conclude the same of the rest of them? So if we could be [19]assur'd in but one Comet, what it was that is the Cause of that

strange Appearance, should we not make that a Standard to judge of all others by?
Arguments from their Similitude, of no small weight. 'Tis therefore an Argument of no small Weight that is fetch'd from Relation and Likeness; and to reason from what we see and are sure of, to what we cannot, is no false Logick. This must be our Method in this Treatise, wherein from the Nature and Circumstances of that Planet which we see before our Eyes, we may guess at those that are farther distant from us.

The Planets are solid, and not without Gravity. And, First, 'tis more than probable that the Bodies of the Planets are solid like that of our Earth, and that they don't want what we call Gravity, that Virtue, which like a Loadstone attracts whatsoever is near the Body to its Center. And that they have such a Quality, their very Figure is a Proof; for their Roundness proceeds only from an equal pressure of all their Parts tending to the same Center. Nay more, we are so skilful now-a-days, as to be able to tell how [20] much more or less the Gravitation in *Jupiter* or *Saturn* is than here; of which Discovery and its Author you may read my *Essay of the Causes of Gravitation*.

But now to carry the Search farther, let us see by what Steps we must rise to the attaining some knowledge in the deeper Secrets concerning the State and Furniture of these new Earths. And, first, how likely is it that they may be stock'd with Plants *Have Animals and Plants.* and Animals as well as we? I suppose no Body will deny but that there's somewhat more of Contrivance, somewhat more wonderful in the Production and Growth of Plants and Animals, than in Lifeless Heaps of inanimate Bodies, be they never so much larger; as Mountains, Rocks, or Seas are. For the Finger of God, and the Wisdom of Divine Providence, is in them much more clearly manifested than in the other. One of *Democritus's* or *Cartes's* Scholars may venture perhaps to give some tolerable Explication of the Appearances in Heaven and Earth, allow him but his Atoms [21] and Motion; but when he comes to Plants and Animals, he'll find himself non-plus'd, and give you no likely account of their Production. For every Thing in them is so exactly adapted to some Design, every part of them so fitted to its proper Use, that they manifest an Infinite Wisdom, and exquisite Knowledge in the Laws of Nature and Geometry, as, to omit those Wonders in Generation, we shall by and by show; and make it an Absurdity even to think of their being thus happily jumbled together by a chance Motion of I don't know what little Particles. Now should we allow the Planets nothing but vast Deserts, lifeless and inanimate Stocks and Stones, and deprive them of all those Creatures that more plainly speak their Divine Architect, we should sink them below the Earth in Beauty and Dignity; a Thing very unreasonable, as I said before.

Well then, we have gain'd the Point thus far, and the Planets may be allowed some Creatures capable of moving themselves, not at all inferior [22] to ours; and these are

Animals. And if this be allowed, it almost necessarily follows, that there must be Herbs *Not to be imagin'd too unlike ours.* for Food for them. And as for the Growth and Nourishment of all these, 'tis no doubt the same with ours, seeing they have the same Sun to warm and enliven them as ours have.

But perhaps some Body may say, we conclude too fast. They will not deny indeed but that there may be Plants and Animals on the Surface of the Planets, that deserve as well to be provided for by their Creator as ours do: but why must they be of the same Kind with ours: Nature seems to love variety in her Works, and may have made them widely different from ours either in their matter or manner of Growth, in their outward Shape, or their inward Contexture; she may have made them such as neither our Understanding nor Imagination can conceive. That's the Thing we shall now examine, and whether it be not more likely that she has not observ'd such a Variety as they talk of. Nature [23] seems most commonly, and in most of her Works, to affect Variety, 'tis true; But they should consider 'tis not the Business of Men to pretend to settle how great this Difference and Variety must be. Nor does it follow, because it may be Infinite, and out of our Comprehension and Reach, that therefore Things in reality are so. For suppose God should have pleased to have made all Things in the rest of the Planets just as he has here, the Inhabitants of those Places (if there are any such) would admire his Wisdom and Contrivance no less than if they were widely different; seeing they can't come to know what's done in the other Planets. Who doubts but that God, if he had pleased, might have made the Animals in *America* and other distant Countries nothing like ours? yet we see he has not done it. They have indeed some difference in their Shape, and 'tis fit they should, to distinguish the Plants and Animals of those Countries from ours, who live on this side the Earth; but even in this Variety there is an Agreement, [24] an exact Correspondence in Figure and Shape, the same ways of Growth, and new Productions, and of continuing their own Kind. Their Animals have Feet and Wings like ours, and like ours have Hearts, Lungs, Guts, and the Parts serving to Generation; whereas all these Things, as well with them as us, might, if it had pleased Infinite Wisdom, have been order'd a very different Way. 'Tis plain then that Nature has not exhibited that Variety in her Works that she could, and therefore we must not allow that Weight to this Argument, as upon the Account of it to make every Thing in the Planets quite different from what is here. 'Tis more probable that all the Difference there is between us and them, springs from the greater or less distance and influence from that Fountain of Heat and Life the Sun; which will cause a Difference not so much in their Form and Shape, as in their Matter and Contexture.

Planets have Water. And as for the Matter whereof the Plants and Animals there consist, tho' [25] it is impossible ever to come to the Knowledge of its Nature, yet this we may venture to assert (there being scarce any Doubt of it) that their Growth and Nourishment proceeds

from some liquid Principle. For all Philosophers agree that there can be no other way of Nutrition; some of the Chief among them having made Water to be the Original of all Things: For whatsoever's dry and without Moisture, is without Motion too; and without Motion, it's impossible there should be any Increase. But the Parts of a Liquid being in continual Motion one with another, and insinuating and twisting themselves into the smallest Places, are thereby very proper and apt to add not themselves only, but whatsoever else they may bring along with them, to the Increase and Growth of Bodies. Thus we see that by the Means of Water the Plants grow, blossom, and bear Fruit; and by the Addition of that only, Stones grow together out of Sand. And there's no doubt but that Metals, Crystals, and Jewels, [26] have the same Method of Production: Tho' in them there has been no opportunity to make the same Observation, as well by reason of their slow Advances, as that they are commonly found far from the Places of their Generation; thrown up I suppose by some Earthquakes, or Convulsions. That the Planets are not without Water, is made not improbable by the late Observations: For about *Jupiter* are observed some Spots of a darker Colour than the rest of his Body, which by their continual change show themselves to be Clouds: For the Spots of *Jupiter* which belong to him, and never remove from him, are quite different from these, being sometimes for a long time not to be seen for these Clouds; and again, when these disappear, showing themselves. And at the going off of these Clouds, some Spots have been taken notice of in him, much brighter than the rest of his Body, which remained but a little while, and then were hid from our Sight. These Monsieur *Caffini* thinks are only the [27] Reflection from the Snow that covers the Tops of the Hills in *Jupiter*: But I should rather think that it is only the Colour of the Earth, which happens to be free from those Clouds that commonly darken it.

Mars too is found not to be without his dark Spots, by means of which he has been observed to turn round his own Axis in 24 Hours and 40 Minutes; the Length of his Day: but whether he has Clouds or no, we have not had the same opportunity of observing as in *Jupiter*, as well because even when he is nearest the Earth, he appears to us much less than *Jupiter*, as that his Light not coming so far, is so brisk as to be an Impediment to exact Observations: And this Reason is as much stronger in *Venus* as its Light is. But since 'tis certain that the Earth and *Jupiter* have their Water and Clouds, there is no Reason why the other Planets *But not just like ours*. should be without them. I can't say that they are exactly of the same nature with our Water; but that they should be liquid their Use requires, as their Beauty does that they should be [28] clear. For this Water of ours, in *Jupiter* or *Saturn*, would be frozen up instantly by reason of the vast distance of the Sun. Every Planet therefore must have its Waters of such a temper, as to be proportioned to its Heat: *Jupiter's* and *Saturn's* must be of such a Nature as not to be liable to Frost; and

Venus's and *Mercury's* of such, as not to be easily evaporated by the Sun. But in all of them, for a continual supply of Moisture, whatever Water is drawn up by the Heat of the Sun into Vapours, must necessarily return back again thither. And this it cannot do but in Drops, which are caused as well there as with us, by their ascending into a higher and colder Region of the Air, out of that which, by reason of the Reflection of the Rays of the Sun from the Earth, is warmer and more temperate.

Here then we have found in these new Worlds Fields warm'd by the kindly Heat of the Sun, and water'd with fruitful Dews and Showers: That there must be Plants in them as well for Ornament as Use, we have shewn [29] just now. And what Nourishment, what manner of Growth shall we allow *Plants grow and are nourished there as they are here.* them? Probably, there can be no better, nay no other, than what we here experience; by having their Roots fastned into the Earth, and imbibing its nourishing Juices by their tender Fibres. And that they may not be only like so many bare Heaths, with nothing but creeping Shrubs and Bushes, we may allow them some nobler and loftier Plants, Trees, or somewhat like them: These being the greatest, and, except Waters, the only Ornament that Nature has bestowed upon the Earth. For not to speak of those many uses that are made of their Wood, there's no one that is ignorant either of their Beauty or Pleasantsness. Now what way can any one imagine for a continual Production and Succession of these Plants, but their bearing Seed? A Method so excellent, that it's the only one that Nature has here made use of, and so wonderful, that it seems to be designed not for this Earth alone. In fine, there's the same reason to think that this Method is observed in those [30] distant Countries, as there was of its being followed in the remote Quarters of this same Earth.

The same true of their Animals. 'Tis much the same in Animals as 'tis in Plants, as to their manner of Nourishment, and Propagation of their Kind. For since all the living Creatures of this Earth, whether Beasts, Birds, Fishes, Worms, or Insects, universally and inviolably follow the same constant and fix'd Institution of Nature; all feed on Herbs, or Fruits, or the Flesh of other Animals that fed on them: since all Generation is performed by the impregnating of the Eggs, and the Copulation of Male and Female: Why may not the same Rule be observed in the Planetary Worlds? For *'tis certain that the Herbs and Animals that are there would be lost, their whole Species destroyed without some daily new Productions:* except there be no such thing there as Misfortune or Accident: except the Plants are not like other humid Bodies, but can bear Heat, Frost, and Age, without being dry'd up, kill'd or decay'd: except the Animals have Bodies as hard [31] and durable as Marble; which I think are gross Absurdities. If we should invent some new Way for their coming into the World, and make them drop like Soland Geese from Trees, how ridiculous would this be to any one that considers the vast Difference between Wood

and Flesh? Or suppose we should have new ones made every Day out of some such fruitful Mud as that of *Nile*, who does not see how contrary this is to all that's reasonable? And that 'tis much more agreeable to the Wisdom of God, once for all to create of all sorts of Animals, and distribute them all over the Earth in such a wonderful and inconceivable way as he has, than to be continually obliged to new Productions out of the Earth? And what miserable, what helpless Creatures must these be, when there's no one that by his Duty will be obliged, or by that strange natural fondness, which God has wisely made a necessary Argument for all Animals to take care of their own, will be moved to assist, nurse or educate them? [32]

As for what I have said concerning their Propagation, I cannot be so positive; but the other Thing, namely, that they have Plants and Animals, I think I have fully proved, viz. from hence, that otherwise they would be inferior to our Earth. And by the same Argument, they must have as great a Variety of both as we have. What this is, will be best known to him that considers the different Ways our Animals make use of in moving from one Place to another. Which may be reduc'd, I think, to these, either that they walk upon two Feet or Four; or like Insects, upon Six, nay sometimes Hundreds; or that they fly in the Air bearing up, and wonderfully steering themselves with their Wings; or creep upon the Ground without Feet; or by a violent Spring in their Bodies, or paddling with their Feet, cut themselves a Way in the Waters. I don't believe, nor can I conceive, that there should be any other Way than these mentioned. The Animals then in the Planets must make use of one or more of these, like our amphibious Birds, [33] which can swim in Water as well as walk on Land, or fly in the Air; or like our Crocodiles and Sea-Horses, must be Mongrels, between Land and Water. There can no other Method be imagined but one of these. For where is it possible for Animals to live, except upon such a solid Body as our Earth, or a fluid one like the Water, or still a more fluid one than that, such as our Air is? The Air I confess may be much thicker and heavier than ours, and so, without any Disadvantage to its Transparency, be fitter for the volatile Animals. There may also be many sorts of Fluids ranged over one another in Rows as it were. The Sea perhaps may have such a fluid lying on it, which tho' ten times lighter than Water, may be a hundred Times heavier than Air; whose utmost Extent may not be so large as to cover the higher Places of their Earth. But there's no Reason to suspect or allow them this, since we have no such Thing; and if we did, it would be of no Advantage to them, for that the former Ways of moving would not be hereby at all increas'd: [34] But when we come to meddle with the Shape of these Creatures, and consider the incredible Variety that is even in those of the different parts of this Earth, and that *America* has some which are nowhere else to be found, I must then confess that I think it beyond the Force of Imagination to arrive at any knowledge in the Matter, or reach to Probability concerning the Figures of

these Planetary Animals. Altho' confidering these Ways of Motion we e'en now recounted, they may perhaps be no more different from ours than ours (thofe of ours I mean that are moft unlike) are from one another.

If a Man were admitted to a Survey of *Jupiter* or *Venus*, he would no doubt find as great a Number and Variety as he had at home. Let us then, that we may make as near a Guefs at, and as reafonable a Judgment of the Matter as we can, confider the many Sorts, and the admirable Difference in the Shapes of our own Animals; running *Great Variety of Animals in this Earth*. over fome of the Chief of them (for 'twould be tedious to fet about a general Catalogue) that are notoriously [35]different from one another, either in the Figure or fome peculiar Property belonging to them; as they belong to the Land, or the Water, or the Air. Among the Beafts we may take notice of the great Diftance between the Horfe, the Elephant, the Lion, the Stag, the Camel, the Hog, the Ape, the Porcupine, the Tortoife, the Cameleon: in the Water, of that between the Whale, and the Sea-Calf, the Skait, the Pike, the Eel, the Ink-Fifh, the Pourcontrol, the Crocodile, the Flying-fifh, the Cramp-fifh, the Crab, the Oifter, and the Purple-Fifh: and among Birds, of that between the Eagle, the Oftrich, the Peacock, the Swan, the Owl, and the Bat: and in Infects, of that between the Ants, the Spider, the Fly, and the Butterfly; and of that Prodigy in their wonderful change from Worms. In this Roll I have pafs'd by the creeping Kind as one Sort, and skip'd over that vaft Multitude of lefs different Animals that fill the intermediate Spaces. But be they never fo many, there is no *And no lefs in the Planets*. reafon to think that the Planets cannot match them. For tho' we in vain guefs [36]at the Figures of thofe Creatures, yet we have difcover'd fomewhat of their manner of Life in general; and of their Senfes we fhall fpeak more by and by.

The fame in Plants. The more confiderable Differences in our Plants ought to be thought on, as well as the other. As in Trees, that between the Fir and the Oak, the Palm, the Vine, the Fig, and the Coco-Nut Tree, and that in the *Indies*, from whole Boughs new Roots fpring, and grow downwards into the Earth. In Herbs, the Difference is notable between Grafs, Poppy, Colewort, Ivy, Pompions, and the Indian Fig with thick Leaves growing up without any Stalk, and Aloe. Between every one of which again there are many lefs differing Plants not taken notice of. Then the different Ways of railing them are remarkable, whether from Seeds, or Kernels, or Roots, or by grafting or inoculating them. And yet in all thefe, whether we confider the Things themfelves, or the Ways of their Production, I make no doubt but that the Planetary Worlds have as wonderful a Variety as we. [37]

But ftill the main and moft agreeable Point of the Enquiry is behind, *Rational Animals in the Planets*. which is the placing fome Spectators in thefe new Difcoveries, to enjoy thefe

Creatures we have planted them with, and to admire their Beauty and Variety. And among all, that have never so lightly meddled with these Matters, I don't find any that have scrupled to allow them their Inhabitants: not Men perhaps like ours, but some Creatures or other endued with Reason. For all this Furniture and Beauty the Planets are stock'd with seem to have been made in vain, without any Design or End, unless there were some in them that might at the same time enjoy the Fruits, and adore the wise Creator of them. But this alone would be no prevailing Argument with me to allow them such Creatures. For what if we should say, that God made them for no other Design, but that he himself might see (not as we do 'tis true; but that he that made the Eye sees, who can doubt?) and delight himself in the Contemplation of them? For was not [38]Man himself, and all that the whole World contains, made upon this very account? That which makes me of this Opinion, that those Worlds are not without such a Creature endued with Reason, is, that otherwise our Earth would have too much the Advantage of them, in being the only part of the Universe that could boast of such a Creature so far above, not only Plants and Trees, but all Animals whatsoever: a Creature that has something Divine in him, that knows, and understands, and remembers such an innumerable number of Things; that deliberates, weighs and judges of the Truth: A Creature upon whose Account, and for whose Use, whatsoever the Earth brings forth seems to be provided. For every Thing here he converts to his own Ends. With the Trees, Stones, and Metals, he builds himself Houses: the Birds and Fishes he sustains himself with: and the Water and Winds he makes subservient to his Navigation, as he doth the sweet Smell and glorious Colours of the Flowers to his Delight. What can there be [39]in the Planets that can make up for its Defects in the want of so noble an Animal? If we should allow *Jupiter* a greater Variety of other Creatures, more Trees, Herbs and Metals, all these would not advantage or dignify that Planet so much as that one Animal doth ours by the admirable Productions of his penetrating Wit. If I am mistaken in this, I do not know when to trust my Reason, and must allow my self to be but a poor Judge in the true Estimate of Things.

Vices of Men no hindrance to their being the Glory of the Planet they inhabit. Nor let any one say here, that there's so much Villany and Wickedness in Man that we have thus magnified, that it's a reasonable Doubt, whether he would not be so far from being the Glory and Ornament of the Planet that enjoys his Company, that he would be rather its Shame and Disgrace. For first, the Vices that most Men are tainted with, are no hindrance, but that those that follow the Dictates of true Reason, and obey the Rules of a rigid Virtue, are still a Beauty and Ornament to the Place that has the Happiness to harbour them. [40]Besides, the Vices of Men themselves are of excellent Use, and are not permitted and allowed in the World without wise Design. For since it has so pleased God to order the Earth, and every Thing in it as we see it is (for it's absurd to say it happen'd against his Will or Knowledge) we

muft not think that fo great a Diverfity of Minds were placed in different Men to no End or Purpofe: but that this mixture of bad Men with Good, and the Confequents of fuch a Mixture, as Misfortunes, Wars, Afflictions, Poverty, and the like, were permitted for this very good End, viz. the exercifing our Wits, and fharpning our Inventions; by forcing us to provide for our own neceffary Defence againft our Enemies. 'Tis to the Fear of Poverty and Mifery that we are beholden for all our Arts, and for that natural Knowledge which was the Product of laborious Induftry; and which makes us that we cannot but admire the Power and Wifdom of the Creator, which otherwife we might have paffed by with the fame indifference as Beafts. And if Men [41] were to lead their whole Lives in an undifturbed continual Peace, in no fear of Poverty, no danger of War, I doubt they would live little better than Brutes, without all knowledge or enjoyment of thofe Advantages that make our Lives pafs on with Pleafure and Profit. We fhould want the wonderful Art of Writing, if its great Ufe and neceffity in Commerce and War had not forced out the Invention. 'Tis to thefe we owe our Art of Sailing, our Art of Sowing, and moft of thofe Discoveries of which we are Mafters; and almoft all the Secrets in experimental Knowledge. So that thofe very Things on account of which the Faculty of Reafon feems to have been accufed, are no fmall helps to its Advancement and Perfection. For thofe Virtues themfelves, Fortitude and Conftancy, would be of no ufe if there were no Dangers, no Adverfity, no Afflictions for their Exercife and Trial.

If we fhould therefore imagine in the Planets fome fuch reasonable Creature as Man is, adorn'd with the fame [42] Virtues, and liable to the fame Vices, it would be fo far from degrading or vilifying them, that while they want fuch a one, I muft think them inferior to our Earth.

Reafon they are not different from what 'tis here. But if we allow thefe Planetary Inhabitants fome fort of Reafon, muft it needs, may fome fay, be the fame with ours? Certainly it muft, whether we confider it as applied to Juftice and Morality, or exercifed in the Principles and Foundations of Science. For Reafon with us is that which gives us a true Senfe of Juftice and Honefty, Praise, Kindnefs and Gratitude: 'tis That that teaches us to diftinguifh univerfally between Good and Bad; and renders us capable of Knowledge and Experience in it. And can there be any where any other Sort of Reafon than this? or can what we call juft and generous, in *Jupiter* or *Mars* be thought unjuft Villany? This is not at all, I don't fay probable, but poffible. For the Aim and Defign of the Creator is every where the Prefervation and Safety of his Creatures. Now when fuch Reafon as we are Mafters [43] of, is neceffary for the prefervation of Life, and promoting of Society (a thing that they are not without, as we fhall fhew) would it not be ftrange that the Planetary Inhabitants fhould have fuch a perverfe Sort of Reafon given them, as would neceffarily deftroy and confound what it was defign'd to maintain and defend? But allowing

Morality and Passions with those distant Inhabitants to be somewhat different from ours, and supposing they may act by other Principles in what belongs to Friendship and Anger, Hatred, Honesty, Modesty, and Comeliness, yet still there would be no doubt, but that in the Search after Truth, in judging of the Consequences of Things, in Reasoning, particularly in that Sort which belongs to Magnitude or Quantity, about which their Geometry (if they have such a Thing) is employ'd, there would be no doubt, I say, but that their Reason here must be exactly the same, and go the same way to work with ours, and that what's true in one part will hold true over the whole Universe; so that all the difference [44] must lie in the Degrees of Knowledge, which will be proportional to the Genius and Capacity of the Inhabitants.

They have Senses. But I perceive I am got somewhat too far: Let us first enquire a little concerning the bodily Senses of these Planetary Persons; for without such, neither will Life be any Pleasure to them, nor Reason of any Use. And I think it very probable, that all their Animals, as well their Beasts as rational Creatures, are like ours in all that relates to the Senses: For without the Power of Seeing we should find it impossible for Animals to provide Food for themselves, or be fore-warn'd of any approaching Danger, so as to guard themselves from it. So that where-ever we plant any Animals, except we wou'd have them lead the Life of Worms or Moles, we must allow them Sight; than which nothing can conduce more either to the Preservation or Pleasure of their Lives. Then if we consider the wonderful Nature of Light, and the *Sight.* amazing Artifice in the fit framing the Eye for the Reception of it, we cannot [45] but see that Bodies so vastly remote could not be perceived by us in their proper Figures and just Distances, any other way than by Sight. For this Sense, and all others that we know of, must proceed from an external Motion. Which in the sense of Seeing must come either from the Sun, the fix'd Stars, or Fire: whose Particles being put into a very quick Motion, communicate it to the Celestial Matter about, whence 'tis convey'd in a very short time to the most distant parts, just like Sound through the Air. If it were not for this Motion of the intermediate Ætherial Matter, we should be all in Darkness, and have Sight neither of Sun nor Stars, nor any thing else, for all other Light must come to us by Reflection from them. This Motion perceived by the Eyes is called Light. And the nice Curiosity of this Perception is admirable, in that it is caused by the smallest Particles of the luminous Body brought to us by that fine Matter, which at the same time determine the Coast from whence the Motion comes; and in that all these different Roads of [46] Motion, these Waves crossing and interfering with one another, are yet no hindrance to every one's free Passage. All these Things are so wisely, so wonderfully contrived, that it's above the Power of humane Wit, to invent or frame any thing like them; nay, it is very difficult so much as to imagine and comprehend them. For what can be more amazing, than that one small Part of the

Body should be so devised and framed, as by its means to show us the Shape, the Position, the Distance, and all the Motions, nay, and all the Colours, of a Body that is far remote from us, that it may appear the more distinct? And then the artful Composition of the Eye, drawing an exact Picture of the Objects without it, upon the concave Side of the Choroides, is even above all Admiration, nor is there any Thing in which God has more plainly manifested his excellent Geometry. And these Things are not only contriv'd and fram'd with so great Wisdom and Skill, as not to admit of better, but to any one that considers them attentively, they seem to be of such a Nature [47]as not to allow any other Method. For it's impossible that Light should represent Objects to us at so vast a distance, except by such an intervening Motion; and it's as impossible that any other Composition of the Eye should be equally fitted to the Reception of such Impressions. So that I cannot but think them greatly mistaken, that maintain these Things might have been contrived many other Ways. It's likely then, and credible, that in these Things the Planets have an exact correspondence with us, and that their Animals have the same Organs, and use the same way of Sight that we do. They must have Eyes therefore, and two at least we must grant them, otherwise they would not perceive those Things close to them, nor hardly be able to walk about with Safety. And if we must allow them to all Animals for the Preservation of their Life, how much more must they that make more, and more noble Uses of them, not be deprived of the Blessing of so advantageous Members? For by them we view the various Flowers, and the elegant Features of Beauty: with [48]them we read, we write, we contemplate the Heavens and Stars, and measure their Distances, Magnitudes, and Journeys: which how far they are common to the Inhabitants of those Worlds with us, I shall presently examine. But first I shall enquire whether now we have given them one, we ought also to give them the other *Hearing*. four Senses. And indeed as to Hearing many Arguments persuade me to give it a Share in the Animals of those new Worlds. For 'tis of great consequence in defending us from sudden Accidents; and, especially when Seeing is of no use to us, it supplies its Place, and gives us seasonable warning of any imminent Danger. Besides, we see many Animals call their Fellow to them with their Voice, which Language may have more in it than we are aware of, tho' we don't understand it. But if we do but consider the vast Uses and necessary Occasions of Speaking on the one side, and Hearing on the other, among those Creatures that make use of their Reason, it will scarce seem credible that two such useful, such [49]excellent Things were designed only for us. For how is it possible but that they that are without these, must be without many other Necessaries and Conveniences of Life? Or what can they have to recompense this Want? Then, if we go still farther, and do but meditate upon the neat and frugal Contrivance of Nature in making the same Air, by the drawing in of which we live, by whose Motion we sail, and by whose Means Birds fly, for a Conveyance of Sound to our Ears; and this Sound for the Conveyance of another Man's Thoughts to our

Minds: Can we ever imagine that she has left those other Worlds destitute of so vast *A Medium to convey Sound to the Ears*. Advantages? That they don't want the Means of them is certain, for their having Clouds in *Jupiter* puts it past doubt that they have Air too; that being mostly formed of the Particles of Water flying about, as the Clouds are of them gathered into small Drops. And another Proof of it is, the necessity of breathing for the preservation of Life, a Thing that seems to be as universal a Dictate of Nature, as feeding upon the Fruits of the Earth. [50]

Touch. As for Feeling, it seems to be given upon necessity to all Creatures that are cover'd with a fine and sensible Skin, as a Caution against coming too near those Things that may injure or incommode them: and without it they would be liable to continual Wounds, Blows and Bruises. Nature seems to have been so sensible of this, that she has not left the least place free from such a Perception. Therefore it's probable that the Inhabitants of those Worlds are not without so necessary a Defence, and so fit a Preservative against Dangers and Mishaps.

Smell and Taste. And who is there that doth not see the inevitable necessity for all Creatures that live by feeding to have both Taste and Smell, that they may distinguish those Things that are good and nourishing, from those that are mischievous and harmful? If therefore we allow the Planetary Creatures to feed upon Herbs, Seeds, or Flesh, we must allow them Taste and Smell, that they may chuse or refuse any Thing according as they find it likely to be advantageous or noxious to them. [51]

I know that it hath been a Question with many, whether there might not have been more Senses than these five. *Their Senses not very different from ours*. If we should allow this, it might nevertheless be reasonably doubted, whether the Senses of the Planetary Inhabitants are much different from ours. I must confess, I cannot deny but there might possibly have been more Senses; but when I consider the Uses of those we have, I cannot think but they would have been superfluous. The Eye was made to discern near and remote Objects, the Ear to give us notice of what our Eyes could not, either in the Dark or behind our Back: Then what neither the Eye nor the Ear could, the Nose was made (which in Dogs is wonderfully nice) to warn us of. And if any thing escapes the notice of the other four Senses, we have Feeling to inform us of the too near Approaches of it before it can do us any mischief. Thus has Nature so plentifully, so perfectly provided for the necessary preservation of her Creatures here, that I think she can give nothing more to those there, but what [52] will be needless and superfluous. Yet the Senses were not wholly designed for use: but Men from all, and all other Animals from some of them, reap Pleasure as well as Profit, as from the Taste in delicious Meats; from the Smell in Flowers and Perfumes; from the Sight in the Contemplation of beautiful Shapes and Colours; from the Hearing

in the Sweetness and Harmony of Sounds; from the Feeling in Copulation, unless you please to count that for a particular Sense by it self. *They have Pleasure arising from the Senses.* Since it is thus, I think 'tis but reasonable to allow the Inhabitants of the Planets these same Advantages that we have from them. For upon this Consideration only, how much happier and easier a Man's Life is rendered by the enjoyment of them, we must be obliged to grant them these Blessings, except we would engross every thing that is good to our selves, as if we were worthier and more deserving than any else. But moreover, that Pleasure which we perceive in Eating or in Copulation, seems to be a necessary and provident Command of Nature, [53] whereby it tacitly compels us to the preservation and continuance of our Life and Kind. It is the same in Beasts. So that both for their Happiness and Preservation it's very probable the rest of the Planets are not without it. Certainly when I consider all these Things, how great, noble, and useful they are; when I consider what an admirable Providence it is that there's such a Thing as Pleasure in the World, I can't but think that our Earth, the smallest part almost of the Universe, was never design'd to monopolize so great a Blessing. And thus much for those Pleasures which affect our bodily Senses, but have little or no relation to our Reason and Mind. But there are other Pleasures which Men enjoy, which their Soul only and Reason can relish: Some airy and brisk, others grave and solid, and yet nevertheless Pleasures, as arising from the Satisfaction which we feel in Knowledge and Inventions, and Searches after Truth, of which whether the Planetary Inhabitants are not partakers, we shall have an opportunity of enquiring by and by. There [54]

There are some other things to be consider'd first, in which it's probable they have some relation to us. That the Planets have those Elements of Earth, Air and Water, as well as we, I have already made not unlikely. Let us now see whether they may not have Fire also: which is not so properly call'd an Element, as a very quick Motion of the Particles in the inflammable *All the Planets have Fire.* Body. But be it what it will, there are many Arguments for their not being without it. For this Earth is not so truly call'd the Place of Fire as the Sun: and as by the Heat of that all Plants and Animals here thrive and live; so, no doubt, it is in the other Planets. Since then Fire is caused by a most intense and vigorous Heat, it follows that the Planets, especially those nearer the Fountain of it, have their proportionate degrees of Heat and Fire. And since there are so many ways of its Production, as by the collection of the Rays of the Sun, by the reflection of Mirrors, by the striking of Flint and Steel, by the rubbing of Wood, by the close loading of [55] moist Grains, by Lightning, by the eruptions of Mountains and Volcanos, it's strange if neither Art should have produced it, nor Nature effected it there by one of these many means. Then how useful and necessary is it to us? By it we drive away Cold, and supply the want of the Sun in those Countries where his oblique Rays make a less vigorous Impression,

and so keep a great part of the Earth from being an uninhabited Desert: which is equally necessary in all the Planets, whether we allow them Succession of Seasons, or a perpetual Spring and Æquinox: for even then the Countries near the Pole would receive but little Advantage from the Heat of the Sun. By the help of this we turn the Night into Day, and thereby make a considerable addition to the shortness of our Lives. Upon all these Accounts we ought not to think this Earth of ours enjoy it all alone, and exclude all the other Planets from so advantageous and so profitable a Gift.

But perhaps it may be asked as well concerning Brutes as rational Creatures, [56]and of their Plants and Trees too, whether they are proportionably *The bigness of their Creatures not rightly quest at by the bigness of the Planets.* larger or less than ours. For if the Magnitude of the Planets was to be the Standard of their measure, there would be Animals in *Jupiter* ten or fifteen times larger than Elephants, and as much longer than our Whales, and then their Men must be all Giants in respect to us. Now tho' I don't see any so great Absurdity in this as to make it impossible, yet there is no reason to think it is really so, seeing Nature has not always ty'd her self to those Rules which we have thought more convenient for her: For example, the Magnitude of the Planets is not answerable to their distances from the Sun; but *Mars*, tho' more remote, is far less than *Venus*: and *Jupiter* turns round his Axis in ten Hours, when the Earth which is much less than him, spends 24. But since Nature, perhaps some will say, has not observed such a Regularity in the proportion of Things, for ought we know there may be only a Race of Pygmies about the Bigness of Frogs and Mice, [57]possess'd of the Planets. But I shall show that this is very improbable by and by.

In the Planets are sorts of rational Creatures as well as here. There may arise another Question, whether there be in the Planets but one sort of rational Creatures, or if there be not several sorts possess'd of different degrees of Reason and Sense. There is something not unlike this to be observed among us. For to pass by those who have human Shape (altho' some of them would very well bear that Enquiry too) if we do but consider some sorts of Beasts, as the Dog, the Ape, the Beaver, the Elephant, nay some Birds and Bees, what Sense and Understanding they are masters of, we shall be forced to allow, that Man is not the only rational Animal. For we discover somewhat in them of Reason independent on, and prior to all Teaching and Practice.

But still no Body can doubt, but that the Understanding and Reason of Man is to be prefer'd to theirs, as being comprehensive of innumerable Things, indued with an infinite memory of what's past, and capable of providing [58]against what's to come. That there is some such Species of rational Creatures in the other Planets, which is the Head and Sovereign of the rest, is very reasonable to believe: for otherwise, were many Species

endued with the same Wisdom and Cunning, we should have them always doing Mischief, always quarrelling and fighting one with another for Empire and Sovereignty, a Thing that we feel too much of where we have but one such Species. But to let that pass, our next Enquiry shall be concerning those Animals in the Planets which are furnished with the greatest Reason, whether it's possible to know wherein they employ it, and whether they have made as great Advances in Arts and Knowledge as we in our Planet. Which deserves most to be considered and examined of any thing belonging to their Nature; and for the better Performance of it we must take our Rise somewhat higher, and nicely view the Lives and Studies of Men.

And in those things wherein Men provide and take care only of what's [59]absolutely necessary for the preservation of their Life; in defending themselves from the Injuries of the Air; in securing themselves against the Incurfions of Enemies by Walls; and against Fraud and Disturbances by Laws; in educating their Children, and providing for themselves and them: In all these I can see no great reason that Man has to boast of the Pre-eminency of his Reason above Beasts and other Animals. For most of these Things they perform with greater Ease and Art than we, and some of them they have no need of. For that Sense of Virtue and Justice in which Man excels, of Friendship, Gratitude and Honesty, of what use are they, but either to put a stop to the Wickedness of Man, or to secure us from mutual Assaults and Injuries, Things wherein the Beasts want no Guide but Nature and Inclination? Then if we set before our Eyes the manifold Cares, the Disturbances of Mind, the restless Desires, the dread of Death, that are the result of this our Reason; and compare them with [60]that easy, quiet, and harmless Life which other Animals enjoy, we should be apt to wish a Change, and conclude that they, especially Birds, lived with more Pleasure and Happiness than Man could with all his Wisdom. For they have as great a Relish of bodily Pleasures as we, let the new Philosophers say what they will, who would have them to be nothing but Clocks and Engines of Flesh; a Thing which Beasts so plainly confute by crying and running away from a Stick, and all other Actions, that I wonder how any one could subscribe to so absurd and cruel an Opinion. Nay, I can scarce doubt but that Birds feel no small Pleasure in their easy, smooth sailing through the Air; and would much more if they but knew the Advantages it hath above our slow and *Men chiefly differ from Beasts in the Study of Nature.* laborious Progression. What is it then after all that sets human Reason above all other, and makes us preferable to the rest of the Animal World? Nothing in my Mind so much as the Contemplation of the Works of God, and the Study of Nature, and the improving [61]those Sciences which may bring us to some knowledge in their Beauty and Variety. For without Knowledge what would be Contemplation? And what difference is there between a Man, who with a careless supine Negligence views the Beauty and Use of the Sun, and the fine golden Furniture of

the Heaven, and one who with a learned Niceneſs ſearches into their Courſes; who underſtands wherein the Fix'd Stars, as they are call'd, differ from the Planets, and what is the Reaſon of the regular Viciffitude of the Seaſons; who by ſound Reaſoning can meaſure the Magnitude and Diſtance of the Sun and Planets? Or between ſuch a one as admires perhaps the nimble Activity and ſtrange Motions of ſome Animals, and one that knows their whole Structure, underſtands the whole Fabrick and Architecture of their Compoſition? If therefore the Principle we before laid down be true, that the other Planets are not inferiour in Dignity to ours, *They have Aſtronomy*. what follows but that they have Creatures not to ſtare and wonder at the [62] Works of Nature only, but who employ their Reaſon in the Examination and Knowledge of them, and have made as great Advances therein as we have? They do not only view the Stars, but they improve the Science of Aſtronomy: nor is there any thing can make us think this improbable, but that fond Conceitedneſs of every Thing that we call our own, and that Pride that is too natural to us to be eaſily laid down. But I know ſome will ſay, we are a little too bold in theſe Aſſertions of the Planets, and that we mounted hither by many Probabilities, one of which, if it chance to be falſe, and contrary to our Suppoſition, would, like a bad Foundation, ruin the whole Building, and make it fall to the Ground. But I would have them to know, that all I have ſaid of their Knowledge in Aſtronomy, has Proofs enough, antecedent to thoſe we now produced. For ſuppoſing the Earth, as we did, one of the Planets of equal Dignity and Honour with the reſt, who would venture to ſay, that no where elſe were to be found any that [63] enjoy'd the glorious Sight of Nature's Theatre? Or if there were any Fellow-Spectators, yet we were the only ones that had dived deep into the Secrets and Knowledge of it? So that here's a Proof not ſo far fetch'd for the Aſtronomy of the Planets, the ſame which we uſed for their having rational Creatures, and enjoying the other Advantages we before talk'd of, which ſerves at the ſame time for the Confirmation of our former Conjectures. But if Amazement and Fear at the Eclipſes of the Moon and Sun gave the firſt occaſion to the Study of Aſtronomy, as probably they did, then it's almoſt impoſſible that *Jupiter* and *Saturn* ſhould be without it; the Argument being of much greater force in them, by reaſon of the daily Eclipſes of their Moons, and the frequent ones of the Sun to their Inhabitants. So that if a Perſon diſintereſted in his Judgment, and equally ignorant of the Affairs of all the Planets, were to give his Opinion in this Matter, I don't doubt he would give the Cauſe for Aſtronomy to thoſe two Planets rather than us. [64]

This Suppoſition of their Knowledge and Uſe of Aſtronomy in the Planetary Worlds, will afford us many new Conjectures about their manner of Life, and their State as to other things.

And all its subservient Arts. For, First: No Observations of the Stars that are necessary to the Knowledge of their Motions, can be made without Instruments; nor can these be made without Metal, Wood, or some such solid Body. Here's a necessity of allowing them the Carpenters Tools, the Saw, the Ax, the Plane, the Mallet, the File: and the making of these requires the Use of Iron, or some equally hard Metal. *Geometry and Arithmetick:* Again, these Instruments can't be without a Circle divided into equal Parts, or a straight Line into unequal. Here's a necessity for introducing Geometry and Arithmetick. Then the Necessity *And Writing.* in such Observations of marking down the Epochas or Accounts of Time, and of transmitting them to Posterity, will force us to grant them the Art of Writing; perhaps very different from ours which is commonly used, but I dare affirm not more ingenious [65] or easy. For how much more ready and expeditious is our Way, than by that multitude of Characters used in *China*; and how vastly preferable to Knots tied in Cords, or the Pictures in use among the barbarous People of *Mexico* and *Peru*? There's no Nation in the World but has some way or other of writing or marking down their Thoughts: So that it's no wonder if the Planetary Inhabitants have been taught it by that great School-mistress Necessity, and apply it to the Study of Astronomy and other Sciences. In Astronomical Matters the Necessity of it is moreover apparent from hence, that the Motion of the Stars is as 'twere to be fancied and guess'd at in different Systems, and these Systems to be continually improved and corrected, as later and more exact Observations shall convince the old ones of Faults: all which can never be deliver'd down to succeeding Generations, unless we make use of Letters and Figures.

But after all these large and liberal Allowances to Them, they will still [66] be behind-hand with us. For we have so certain a Knowledge of the true *And Opticks.* System and Frame of the Universe; we have so admirable an Invention of Telescopes to help our failing Eye-sight in the view of the Bigness and different Forms of the Planetary Bodies, in the discovery of the Mountains, and the Shadows of them on the Surface of the Moon, in the bringing to light an innumerable multitude of Stars otherwise invisible, that we must necessarily be far their Masters in that Knowledge. Hence it is almost necessary (except we have a Mind to flatter and complement our selves as the only People that have the Advantage of such excellent Inventions) either to allow the Planetary Inhabitants such sharp Eyes as not to need them, or else the use of Glasses to help the Deficiency of their Sight. And yet I dare not assert this, lest any one should be so disturbed at the Extravagancy of such an Opinion, as to take the measure of my other Conjectures by it, and hiss them all off, upon the account of this alone. [67]

These Sciences not contrary to Nature. But some Body may perhaps object, and that not without reason at first sight, that the Planetary Inhabitants it's likely are destitute of all refined Knowledge, just as the *Americans* were before they had Commerce with the *Europeans*.

For if one confiders the Ignorance of thofe Nations, and of others in *Afia* and *Africa* equally barbarous, it will appear as if the main Defign of the Creator in placing Men upon the Earth was that they might live, and, in a juft fenfe of all the Bleffings and Pleafure they enjoy, worfhip the Fountain of their Happinefs; but that fome few went beyond the Bounds of Nature in their Enquiries after Knowledge. There does not want an Anfwer to thefe Men. For God could not but forefee the Advances Men would make, in their enquiring into the Heavenly Bodies: that they would difcover Arts ufeul and advantageous to Life: that they would crofs the Seas, and dig up the Bowels of the Earth. Nothing of all this could happen contrary to the Mind and Knowledge of the Infinite Author of all Things. And if [68]he forefaw thefe Things would be, he fo appointed and deftin'd them to humane kind. And the Studies of Arts and Sciences cannot be laid to be contrary to Nature, fince in the fearch thereof they are employ'd: efpecially if we confider how great the natural, defire and love of Knowledge, rooted in all Men is. For it's impoffible this fhould have been given them upon no Defign or Account. But they will urge, that if fuch a Knowledge is natural, if we were born for it, why are there fo very few, efpecially in Aftronomy, that profecute thefe Studies? For *Europe* is the only Quarter of the Earth in which there have been any Advancements made in Aftronomy. And as for the Judicial Aftrology, which pretends to foretel what is to come, it is fuch a wretched and oftentimes mifchievous piece of Madnefs, that I do not think it ought to be fo much as named here. And even in *Europe*, not one in a hundred Thoufand meddles with thefe Studies. Befides, its Original and Rife is fo late, that many Ages were paff before [69]the very firft Rudiments of Aftronomy or Geometry (which is neceffary to the learning of it) were known. For every Body is acquainted almoft with its firft Beginnings in *Egypt* and *Greece*. Add to this, that 'tis not yet above fourfcore Years fince the bungling Epicycles were difcarded, and the true and eafy plain Motion of the Planets was difcovered. For the Satisfaction of thefe Scruples, to what we laid before, concerning the Fore-knowledge of God, may be added this; That God never defigned we fhould come into the World Aftronomers or Philofophers; thefe Arts are not infufed into us at our Birth, but were ordered, in long Tracts of Time, by degrees to be the Rewards and Refult of laborious Diligence; efpecially thofe Sciences which are now in debate, are fo much the more difficult and abftrufe, that their late Invention and flow Progreff are fo far from being a Wonder, that it is rather ftrange they were ever difcover'd at all. There are but few, I acknowledge one or two perhaps in an Age, that purfue them, [70]or think them their Bufinefs: but their Number will be very confiderable if we take in thofe that have lived in all the Ages in which Aftronomy hath flourifhed: and no Body can deny them that Happinefs and Contentment which they have pretended to above all others. In fine, it was fufficient that fo fmall a Number fhould make it their Study, fo that the Profit and Advantage of their Inventions might but fpread it felf over all the World. Since then the

Inhabitants of this Earth, let them be never so few, have had Parts and Genius sufficient for the Attainment of this Knowledge; and there's no reason to think the Planetary Inhabitants less ingenious or happy than our selves; we have gain'd our Point, and 'tis probable that they are as skilful Astronomers as we can pretend to be. So that now we may venture to deduce some Consequences from such a Supposition.

We have before shew'd the necessary Dependence and Connexion, not only of Geometry and Arithmetick, but of Mechanical Arts and Instruments with this Science. This leads [71]us naturally to the Enquiry how they can use these Instruments and Engines for the Observation of the Stars, how they can write down such their Observations, and perform other Things which we do with our Hands. So that we must necessarily give them *They have Hands*. Hands, or some other Member, as convenient for all those Uses, instead of them. One of the ancient Philosophers laid such Strefs upon the Use and Conveniency of the Hands, that he made no scruple to affirm, they were the Cause and Foundation of all our Knowledge. By which, I suppose, he meant no more, than that without their Help and Assistance Men could never arrive to the Improvement of their Minds in natural Knowledge: And indeed not without Reason. For suppose instead of them they had had Hoofs like Horses or Bullocks given them, they might have laid indeed the Model and Design of Cities and Houses in their Head, but they would never have been able to have built them. They would have had no Subject of Discourse but what belong'd to their [72]Victuals, Marriages, or Self-preservation. They would have been void of all Knowledge and Memory, and indeed would have been but one degree distant from brute Beasts. What could we invent or imagine that could be so exactly accommodated to all the design'd Uses as the Hands are? Elephants can lay hold of, or throw any thing with their Proboscis, can take up even the smallest Things from the Ground, and can perform such surprising Things with it, that it has not very improperly been call'd their Hand, tho' indeed it is nothing but a Nose somewhat longer than ordinary. Nor do Birds show less Art and Design in the Use of their Bills in the picking up their Meat, and the wonderful Composure of their Nests. But all this is nothing to those Conveniences the Hand is so admirably suited to; nothing to that amazing Contrivance in its Capacity of being stretched, or contracted, or turned to any Part as Occasion shall require. And then, to pass by that nice Sense that the Ends of the Fingers are endued with, even to the feeling and [73]distinguishing most sorts of Bodies in the Dark, what Wisdom and Art is shew'd in the Disposition of the Thumb and Fingers, so as to take up or keep fast hold of any Thing we please? Either then the Planetary Inhabitants must have Hands, or somewhat equally convenient, which it is not easy to conceive; or else we must say that Nature has been kinder not only to us, but even to Squirrels and Monkeys than them.

And Feet. That they have Feet also scarce any one can doubt, that does but consider what we said but just now of Animals different Ways of going along, which it's hard to imagine can be perform'd any other ways than what we there recounted. And of all those, there's none can agree so well with the state of the Planetary Inhabitants, as that that we here make use of. Except (what is not very probable, if they live in Society, as I shall show they do) they have found out the Art of flying in some of those Worlds.

That they are upright. The Stature and Shape of Men here does show forth the Divine Providence [74] so much in its being so fitly adapted to its design'd Uses, that it is not without reason that all the Philosophers have taken notice of it, nor without Probability that the Planetary Inhabitants have their Eyes and Countenance upright, like us, for the more convenient and easy Contemplation and Observations of the Stars. For if the Wisdom of the Creator is so observable, so Praise-worthy in the Position of the other Members; in the convenient Situation of the Eyes, as Watches in the higher Region of the Body; in the removing of the more uncomely Parts out of sight as 'twere, we cannot but think he has almost observed the same Method in the Bodies of those remote Inhabitants. Nor *It follows not therefore that they have the same Shape with us.* does it follow from hence that they must be of the same Shape with us. For there is such an infinite possible variety of Figures to be imagined, that both the Structure of their whole Bodies, and every part of them, both outside and inside, may be quite different from ours. How warmly and conveniently are some Creatures cloth'd with [75] Wool, and how finely are others decked and adorn'd with Feathers? Perhaps among the rational Creatures in the Planets there may some such distinction be observ'd in their Garb and Covering; a Thing in which Beasts seem to excel Men in here. Unless perhaps Men are born naked, for this reason to put them upon employing and exercising their Wits, in the inventing and making that Attire that Nature had made necessary for them. And 'tis this Necessity that has been the greatest, if not only occasion of all the Trade and Commerce, of all the Mechanical Inventions and Discoveries that we are Masters of. Besides, Nature might have another great Conveniency in her Eye, by bringing Men into the World naked, namely, that they might accommodate themselves to all places of the World, and go thicker or thinner cloth'd, according as the Season and Climate they liv'd in requir'd. There may still be conceived a greater difference between us and the Inhabitants of the Planets; for there are some sort of Animals, such [76] as Oysters, Lobsters, and Crab-fish, whose Flesh is on the inside of their Bones as 'twere. But that which hinders me from ascribing such a kind of Frame and Composition to the Planetary Inhabitants, is that Nature seems to have done it only in a few of the meanest Sort of Creatures, and that hereby they would be deprived of that quick easy motion of their Hands and Fingers,

which is so useful and necessary to them, otherwise I should not be much affected with the odd Shape and Figure.

A rational Soul may inhabit another Shape than ours. For 'tis a very ridiculous Opinion, that the common People have got, that 'tis impossible a rational Soul should dwell in any other Shape than ours. And yet as silly as 'tis, it has been the occasion of many Philosophers allowing the Gods no other Shape; nay, the Foundation of a Sect among the Christians, that from hence have the Name of *Anthropomorphites*. This can proceed from nothing but the Weakness, Ignorance, and Prejudice of Men; the same as that other concerning humane Shape, that it is [77] the handsomest and most excellent of all others, when indeed it's nothing but a being accustomed to that Figure that makes us think so, and a Conceit that we and all other Animals naturally have, that no Shape or Colour can be so good as our own. Yet so powerful are these, that were we to meet with a Creature of a much different Shape from Man, with Reason and Speech, we should be much surpris'd and shock'd at the Sight. For if we try to imagine or paint a Creature like a Man in every Thing else, but that has a Neck four times as long, and great round Eyes five or six times as big, and farther distant, we cannot look upon't without the utmost Aversion, altho' at the same time we can give no account of our Dislike.

The Planetarians not less than we. When I just now mentioned the Stature of the Planetary Inhabitants, I hinted that 'twas improbable they should be less than we are. For it's likely, that as our Bodies are made in such a proportion to our Earth, as to render us capable of travelling about it, and making Observations [78] upon its Bulk and Figure, the same Order is observ'd in the Inhabitants of the other Planets, unless in this Particular also, which is very considerable, we would prefer our selves to all others. Then seeing we have before allow'd them Astronomy and Observations, we must give them Bodies and Strength sufficient for the ruling their Instruments, and the erecting their Tubes and Engines. And for this the larger they are the better. For if we should suppose them Dwarfs not above the Bigness of Rats or Mice, they could neither make such Observations as are requisite; nor such Instruments as are necessary to those Observations. Therefore we must suppose them larger than, or at least equal to, our selves, especially in *Jupiter* and *Saturn*, which are so vastly bigger than the Planet which we inhabit.

They live in Society. Astronomy, we said before, could never subsist without the writing down the Observations: Nor could the Art of Writing (any more than the Arts of Carpenters and Founders) ever be found out except in a Society [79] of reasonable Creatures, where the Necessities of Life forc'd them upon Invention: So that it follows from hence, (as was before said) that the Planetary Inhabitants must in this be like us, that they maintain a Society and Fellowship with, and afford mutual Assistances and Helps to one another.

Hereupon we must allow them a settled, not a wandering *Scythian* way of living, as more convenient for Men in such Circumstances. But what follows from hence? Must they not have every thing else proper for such a manner of living granted them too? Must they not have their Governours, Houses, Cities, Trade and Bartering? Why should they not, when even the barbarous People of *America* and other Places were at their first Discovery found to have somewhat of that nature in use among them. I don't say, that Things must be the same there as they are here. We have many that may very well be spared among rational Creatures, and were design'd only for the preservation of Society from all Injury, and for the curbing of those [80]Men who make an ill use of their Reason to the Detriment of others. Perhaps in the Planets they have such plenty and affluence of all good Things, as they neither need or desire to steal from one another; perhaps they may be so just and good as to be at perpetual Peace, and never to lie in wait for, or take away the Life of their Neighbour: perhaps they may not know what Anger or Hatred are; and if so, they must be much happier than we. But it's more likely they have such a mixture of Good with Bad, of Wise with Fools, of War with Peace, and want not that School-mistress of Arts Poverty. For, as was before shown, some good use may be made of these things, but if not, there is no Reason why we should prefer their Condition to our own.

They enjoy the Pleasures of Society. What I am now going to say may seem somewhat more bold, and yet is not less likely than the former. For if these Nations in the Planets live in Society, as I have pretty well show'd they do, 'tis somewhat more than probable that they enjoy not only the [81]Profit, but the Pleasures arising from Society: such as Conversation, Amours, Jestings, and Shews. Otherwise we should make them live without Diversion or Merriment; we should deprive them of the great Sweetness of Life, which it can't well be without, and give our selves such an Advantage over them as Reason will by no means admit of.

But to proceed to a farther Enquiry into their Business and Employment, let's consider what we have not yet mention'd, wherein they may bear any Likeness to us. And first we have good Reason to believe they build themselves Houses, because we are sure they are not without their Showers. For in *Jupiter* have been observed Clouds, big no doubt with Vapours and Water, which hath been proved by many other Arguments, not to be wanting in that Planet. They have Rain then, for otherwise how could all the Vapours drawn up by the Heat of the Sun be disposed of? And Winds, for they are caused only by Vapours dissolved by Heat, and it's [82]plain that they blow in *Jupiter* by the continual Motion and Variety of the *They have Houses to secure 'em from Weather.* Clouds about him. To protect themselves from these, and that they may pass their Nights in Quiet and Safety, they must build themselves Tents or Huts, or live in Holes of the Earth. But why may we not suppose the Planetary Inhabitants to be as good Architects, have as noble Houses, and

as stately Palaces as our selves? Unless we think that every Thing which belongs to our selves is the most beautiful and perfect that can be. And who are we, but a few that live in a little Corner of the World, upon a Ball ten Thousand times less than *Jupiter* or *Saturn*? And yet we must be the only skilful People at Building; and all others must be our Inferiours in the Knowledge of uniform Symetry; and not be able to raise Towers and Pyramids as high, magnificent, and beautiful, as our selves. For my part, I see no reason why they may not be as great Masters as we are, and have the Use of all those Arts subservient to it, as Stone-cutting and Brick-making, [83] and whatsoever else is necessary for it, as Iron, Lead and Glass; or ornamental to it, as Gilding and Picture.

If their Globe is divided like ours, into Sea and Land, as it's evident it is (else whence could all those Vapours in *Jupiter* proceed?) we have great Reason to allow them the Art of Navigation, and not vainly ingross so great, so useful a Thing to our selves. Especially considering the great Advantages *Jupiter* and *Saturn* have for Sailing, in having so many Moons to direct their Course, by whose Guidance they may attain easily to the Knowledge that we are not Masters of, of the Longitude of Places. And what a Multitude of other Things follow from this Allowance? If they have Ships, they must have Sails and Anchors, Ropes, Pullies, and Rudders, which are of particular Use in directing a Ship's Course against the Wind, and in sailing different Ways with the same Gale. And perhaps they may not be without the Use of the Compass too, for the magnetical Matter, which continually passes thro' the Pores of our Earth, [84] is of such a Nature, that it's very probable the Planets have something like *They have Navigation, and all Arts subservient*. it. But there's no doubt but that they must have the Mechanical Arts and Astronomy, without which Navigation can no more subsist, than they can without Geometry.

But Geometry stands in no need of being prov'd after this manner. Nor doth it want Assistance from other Arts which depend upon it, but we may have a nearer and shorter Assurance of their not being without it in those Earths. For that Science is of such singular Worth and Dignity, so peculiarly employs the Understanding, and gives it such a full Comprehension and infallible *As Geometry*. certainty of Truth, as no other Knowledge can pretend to: it is moreover of such a Nature, that its Principles and Foundations must be so immutably the same in all Times and Places, that we cannot without Injustice pretend to monopolize it, and rob the rest of the Universe of such an incomparable Study. Nay Nature it self invites us to be Geometricians; it presents us with Geometrical [85] Figures, with Circles and Squares, with Triangles, Polygons, and Spheres, and proposes them as it were to our Consideration and Study, which abstracting from its Usefulness, is most delightful and ravishing. Who can read *Euclid*, or *Apollonius*, about the Circle, without Admiration? Or *Archimedes* of the Surface of the Sphere, and Quadrature of the Parabola without Amazement? or consider the late ingenious Discoveries of the Moderns

with Boldness and Unconcernedness? And all these Truths are as naked and open, and depend upon the same plain Principles and Axioms in *Jupiter* and *Saturn* as here, which makes it not improbable that there are in the Planets some who partake with us in these delightful and pleasant Studies. But what's the greatest Argument with me, that there are such, is their Use, I had almost said Necessity, in most Affairs of humane Life. Now we are got thus far, what if we should venture somewhat farther, and say, that they have our Inventions of the Tables of Sines, of Logarithms, and [86]Algebra? I know it would sound very odd, and perhaps a little ridiculous, and yet there's no reason but the thinking our selves better than all the World, to hinder them from being as happy in their Discoveries, and as ingenious in their Inventions as we our selves are.

They have Musick. It's the same with Musick as with Geometry, it's every where immutably the same, and always will be so. For all Harmony consists in Concord, and Concord is all the World over fix'd according to the same invariable Measure and Proportion. So that in all Nations the Difference and Distance of Notes is the same, whether they be in a continued gradual Progression, or the Voice makes skips over one to the next. Nay very credible Authors report, that there's a sort of Bird in *America*, that can plainly sing in order fix musical Notes: Whence it follows, that the Laws of Musick are unchangeably fix'd by Nature, and therefore the same Reason holds for their Musick, as we e'en now shew'd for their Geometry. For why, supposing other [87]Nations and Creatures, endued with Reason and Sense as well as we, should not they reap the Pleasures arising from these Senses as well as we too? I don't know what Effect this Argument, from the immutable Nature of these Arts, may have upon the Minds of others; I think it no inconsiderable or contemptible one, but of as great Strength as that which I made use of above to prove that the Planetary Inhabitants had the Sense of Seeing.

But if they take delight in Harmony, there is no doubt but that they have invented Musical Instruments. For they could scarce help lighting upon some or other by chance; the Sound of a tight String, the Noise of the Winds, or the whistling of Reeds, might have given them the hint. From these small Beginnings they perhaps, as well as we, have advanced by degrees to the Use of the Lute, Harp, Flute, and many string'd Instruments. But altho' the Tones are certain and determinate, yet we find among different Nations a quite different manner and rule for Singing; as [88]formerly among the *Dorians*, *Phrygians*, and *Lydians*, and in our Time among the *French*, *Italians*, and *Persians*. In like manner it may so happen, that the Musick of the Inhabitants of the Planets may widely differ from all these, and yet be very good. But why we should look upon their Musick to be worse than ours, there's no reason can be given; neither can we well presume that they want the Use of Half-Notes and Quarter-Notes, seeing the Invention of Half-Notes is so obvious, and the Use of them so agreeable to Nature. Nay, to go a Step

farther, what if they should excel us in the Theory and practick part of Musick, and outdo us in Comforts of vocal and instrumental Musick, so artificially compos'd, that they shew their Skill by the Mixtures of Discords and Concorde? and of this last sort 'tis very likely the 5th and 3d are in use with them.

This is a very bold Assertion, but it may be true for ought we know, and the Inhabitants of the Planets may possibly have a greater insight into the Theory of Musick than has yet been [89]discover'd among us. For if you ask any of our Musicians, why two or more perfect Fifths cannot be used regularly in Composition; some say 'tis to avoid that Sweetness and Lushiousness which arises from the Repetition of this pleasing Chord. Others say, this must be avoided for the sake of that Variety of Chords that are requisite to make a good Composition; and these Reasons are brought by *Cartes* and others. But an Inhabitant of *Jupiter* or *Venus* will perhaps give you a better Reason for this, viz. because when you pass from one perfect Fifth to another, there is such a Change made as immediately alters your Key, you are got into a new Key before the Ear is prepared for it, and the more perfect Chords you use of the same kind in Consecution, by so much the more you offend the Ear by these abrupt Changes.

Again, one of these Inhabitants perhaps can show how it comes about, that in a Song of one or more Parts, the Key cannot be kept so well in the same agreeable Tenour, unless the intermediate Closes and Intervals be so temper'd, [90]as to vary from their usual Proportions, and thereby to bear a little this way or that, in order to regulate the Scale. And why this Temperature is best in the System of the Strings, when out of the Fifth the fourth Part of a Comma is usually cut off; This same thing I have formerly shew'd at large.

But for the regulating the Tone of the Voice (as I before hinted) that may admit of a more easy proof, and we shall give you an Essay of it, since I have mentioned a thing that is not mere Imagination only: I say therefore, if any Person strike those Sounds which the Musicians distinguish by these Letters, C, F, D, G, C, by these agreeable Intervals, altogether perfect, interchangeable, ascending and descending with the Voice: Now this latter sound C will be one Comma, or very small portion lower than the first sounding of C. Because of these perfect Intervals, which are as 4 to 3, 5 to 6, 4 to 3, 2 to 3, an account is made in such a Proportion, as 160 to 162. that is, as 80 to 81, which is what they call a Comma. So that if the same Sound [91]should be repeated nine times, the Voice would fall near the Matter a greater Tone, whose proportion is as 8 to 9. But this the Sense of the Ears by no means endures, but remembers the first Tone, and returns to it again. Therefore we are compell'd to use an occult Temperament, and to sing these imperfect Intervals, from doing which less Offence arises. And for the most part, all Singing wants

this Temperament, as may be collected by the aforefaid Computations. And thefe things we have offer'd to thofe that have fome Knowledge in Geometry.

We have fpoke of thefe Arts and Inventions, which it is very probable the Inhabitants of the Planets partake of in common with us, befides which it feems requifite to take in many other Things that ferve either for the Ufe or Pleafure of their Lives. But what thefe Things are we fhall the better account for, by laying before us many of thofe Things which are found among us. I have before mention'd the Variety of Animals and Vegetables, which very much differ from each other, [92]among which there are fome that differ but little; and I have faid, that there are no lefs differences in thefe Things in the Planetary Worlds.

I fhall now take a fhort view of the Benefits we receive both from thofe Herbs and Animals, and fee whether we may not with very good reafon conclude that the Planetary Inhabitants reap as great and as many from thofe that their Countries afford them.

And here it may be worth our while to take a Review of the Variety and Multitude of our Riches. For Trees and Herbs do not only ferve us for Food, they in their delicious Fruits, thefe in their Seeds, Leaves and Roots; but Herbs moreover furnifh us with Phyfick, and Trees with Timber for our Houfes and Ships. Flax, by the means of thofe two ufeul Arts of Spinning and Weaving, affords us Clothing. Of Hemp or Matweed we twift our felves Thread and fmall Ropes, the former of which we employ in Sails and Nets, the latter in making larger Ropes for Mafts and Anchors. With the fweet Smells and [93]*The Advantages we reap from Herbs and Animals.* beauteous Colours of Flowers we feaft our Senfes: and even thofe of them, that offend our Noftils, or are mifchievous to our Bodies, are feldom without excellent Ufes: or were made perhaps by Nature as a Foil to fet off, and make us the more value the Good by comparing them with thefe. What vaft Advantages and Profit do we reap from the Animals? The Sheep give us Clothing, and the Cows afford us Milk: and both of them their Flefh for our Sufenance. Affes, Camels, and Horfes do, what if we wanted them we muft do our felves, carry our Burdens; and the laft of them we make ufe of, either themfelves to carry us, or in our Coaches to draw us. In which we have fo excellent, fo ufeul an Invention of Wheels, that I can't fuppose the Planets to enjoy Society and all its Confequences, and be without them. Whether they are Pythagoreans there, or feed upon Flefh as we do, I dare not affirm any Thing. Tho' it feems to be allowed Men to feed upon whatfoever may afford them Nourifhment, either on Land, or in [94]Water, upon Herbs, and Pomes, Milk, Eggs, Honey, Filh, and no lefs upon the Flefh of many Birds and Beafes. But it is a furprifing thing! that a rational Creature fhould live upon the Ruin and Deftruction of fuch a number of other his Fellow-Creatures! And yet it does not feem at all unnatural, fince not only he, but even Lions, Wolves, and other

ravenous Beasts, prey upon Flocks of other harmless Things, and make mere Fodder of them; as Eagles do of Pidgeons and Hares; and large Fish of the helpless little ones. We have different sorts of Dogs for Hunting, and what our own Legs cannot, that their Nose and Legs can help us to. But the Use and Profit of Herbs and Animals are not the only Things they are good for, but they raise our Delight and Admiration when we consider their various Forms and Natures, and enquire into all their different ways of Generation: Things so infinitely multifarious, and so delightfully amazing, that the Books of natural Philosophers are deservedly filled with their Encomiums. For even in the [95]very Insects, who can but admire the six-corner'd Cells of the Bees, or the artificial Web of a Spider, or the fine Bag of a Silk-worm, which last affords us, with the Help of incredible Industry, even Shiploads of soft delicate Clothing. This is a short Summary of those many profitable Advantages the animal and herbal World serve us with.

But this is not all. The Bowels of the Earth likewise contribute much to Man's Happiness. For what Art and Cunning does he employ in finding, in digging, in trying Metals, and in melting, refining, and tempering them? *And from Metals.* What Skill and Nicety in beating, drawing or dissolving Gold, so as with inconsiderable Changes to make every Thing he pleases put on that noble Lustre? Of how many and admirable Uses is Iron? and how ignorant in all Mechanical Knowledge were those Nations that were not acquainted with it, so as to have no other Arms but Bows, Clubs, and Spears, made of Wood. There's one Thing indeed we have, which it's a Question whether it has [96]done more harm or good, and that is Gun-powder made of Nitre and Brimstone. At first indeed it seem'd as if we had got a more secure Defense than former Ages against all Assaults, and could easily guard our Towns, by the wonderful Strength of that Invention, against all hostile Invasions: but now we find it has rather encouraged them, and at the same time been no small Occasion of the Decay of Valour, by rendring it and Strength almost useless in War. Had the *Grecian* Emperor who said, *Virtue was ruin'd* only when Slings and Rams first came into use, liv'd in our Days, he might well have complain'd; especially of Bombs, against which neither Art nor Nature is of sufficient Proof: but which lays every Thing, Castles and Towers, be they never so strong, even with the Ground. If for nothing else, yet upon this one account, I think we had better have been without the Discovery. Yet, when we were talking of our Discoveries, it was not to be pass'd over, for the Planets too may have their mischievous as well as useful Inventions. [97]

We are happier in the Uses for which the Air and Water serves us; both of which helps us in our Navigation, and furnishes us with a Strength sufficient, without any Labour of our own, to turn round our Mills and Engines; Things which are of use to us in so many different Employments. For with them we grind our Corn, and squeeze out our Oil; with them we cut Wood, and mill Cloth, and with them we beat our Stuff for Paper. An

incomparable Invention! Where the nastieft ufelefs Scraps of Linen are made to produce fine white Sheets. To these we may add the late discovery of Printing, which not only preserves from Death Arts and Knowledge, but makes them much easier to be attained than before. Nor must we forget the Arts of Engraving and Painting, which from mean Beginnings have improved to that Excellence, that nothing that ever sprung from the Wit of Man can claim Pre-eminence to them. Nor is the way of melting and blowing Glasses, and of polishing and spreading Quick-silver [98] over Looking-Glasses, unworthy of being mentioned, nor above all, the admirable uses that Glasses have been put to in natural Knowledge, since the Invention of the Telescope and Microscope. And no less nice and fine is the Art of making Clocks, some of which are so small as to be no weight to the Bearer; and others so exact as to measure out the Time in as small Portions as any one can desire: the Improvement of both which the World owes to my ¹Inventions.

From the discoveries of our Age. I might add much here of the late Discoveries, most of them of this Age, which have been made in all sorts of Natural Knowledge as well as in Geometry and Astronomy, as of the Weight and Spring of the Air, of the Chymical Experiments that have shown us a way of making Liquors that shall shine in the Dark, and with gentle moving shall burn of themselves. I might mention the Circulation of the Blood through the Veins and Arteries, which was understood indeed before; but now, by the help of the Microscope, has an ocular demonstration [99] in the Tails of some Fishes: of the Generation of Animals, which now is found to be performed no otherwise than by the Seed of one of the same kind; and that in the Seed of the Male are discovered, by the help of Glasses, Millions of sprightly little Animals, which it's probable are the very Offspring of the Animals themselves: a surprising thing, and never before now known!

The Planets have, tho' not these fame, yet as useful Inventions. Thus have I put together all these late Discoveries of our Earth: and now, tho' perhaps some of them may be common to the Planetary Inhabitants with us, yet that they should have all of them is not credible. But then they may have somewhat to make up that Defect, others as good and as useful, and as wonderful, that we want. We have allow'd that they may have rational Creatures among them, and Geometricians, and Musicians: We have prov'd that they live in Societies, have Hands and Feet, are guarded with Houses and Walls: Wherefore if a Man could be carried thither by some powerful Genius, some *Mercury*, I don't [100] doubt 'twould be a very curious sight, curious beyond all Imagination, to see the odd ways, and the unusual manner of their setting about any thing, and their strange methods of living. But since there's no hopes of our going such a Journey, we must be contented with what's in our Power: we may suppose our selves there, and inquire as far as we can into the Astronomy of each Planet, and see in what manner the Heavens present themselves to their Inhabitants. We shall make some Observations of the Eminence of each of them, in

respect of their Magnitude, and number of Moons they have to wait on them; and fhall propofe a new Method of coming to fome Knowledge of the incredible diftance of the fix'd Stars. But firft after this long and deep Thoughtfulnefs we will give our felves a little Reft, and fo put an end to this Book. [\[101\]](#)

¹ *The Author invented the Pendulum for Clocks.* [↑](#)

New Conjectures concerning the Planetary Worlds.

BOOK the Second.

'Twas a pretty many Years ago that I chanc'd to light upon *Athanaſius Kircher's* Book, call'd *The Ecſtatick Journey*, which treats of the nature of the Stars, and of the Things that are to be found in the Superficies of the Planets: I wondered to ſee nothing there of what I had often thought not improbable, but quite other Things, nothing but a Heap of idle unreaſonable Stuff: which I was the more confirm'd in, when, after the writing of the former part, I ran over the Book again. And I thought mine were very conſiderable and weighty Matters if compar'd with *Kircher's*. That other People may be ſatisfied in this, and ſee how vainly thoſe, who caſt off the only Foundations of Probability in ſuch Matters, which we have all the way made uſe of, pretend to philoſophize [102] in this caſe, I think it will not be beſide the Purpoſe to beſtow ſome few Reflections upon that Book.

Kircher's Journey in Ecſtacy examin'd. That ingenious Man ſuppoſing himſelf carried by ſome Angel thro' the vaſt Spaces of Heaven, and round the Stars, tells us, he ſaw a great many things, ſome of which he had out of the Books of Aſtronomers, the reſt are the Product of his own Fancy and Thoughts. But, before he enters upon his Journey, he lays down theſe two Things as certain; that no Motion muſt be allowed the Earth, and that God has made nothing in the Planets, no not ſo much as Herbs, which has either Life or Senſe in it. Leaving then the Syſtem of *Copernicus*, he chuſes *Tycho* for his Guide. But when he ſuppoſes all the fix'd Stars to be Suns, and round each of them places their Planets, here (againſt his Will I ſuppoſe) he has unawares made an infinite number of *Copernican* Syſtems. All which, beſide their own Motion, he abſurdly makes to be carried, with an incredible ſwiftness, in twenty four Hours round the Earth. Since moſt [103] of theſe Worlds are out of the Reach of any Man's light, as he owns they are, I cannot think for what purpoſe he makes ſo many Suns to ſhine upon deſolate Lands (like our Earth in every thing, he ſays, only that they have neither Plants nor Animals) where there's no one to whom they ſhould give light. And from hence he ſtill falls into more and more Abſurdities. And becauſe he could find no other uſe of the Planets, even in our Syſtem, he is forc'd to beg Help of the Aſtrologers; and would have all thoſe vaſt Bodies made upon no other account than that the whole Univerſe might be preſerved and continue ſecure by their means, and that they might govern the Mind of Man by their various and regular Influences. Accordingly, to gratify Aſtrology, he ſays that *Venus* was the moſt pleaſant Place, every thing fine and handſome, its Light gentle, its Waters ſweet and purling, and it ſelf beſet all about with ſhining Chryſtals. In *Jupiter* he found wholeſome and ſweet Gales, delicate Waters, and a Land ſhining like Silver. For from [104] theſe two

Planets it seems, Men have all that is happy and healthful poured down upon them; and all that renders them handsome and lovely, wise and grave, is owing to their Influences. *Mercury* had I don't know what Airiness and Briskness in it; whence Men derive, when they are first born, all their Wit and Cunning. *Mars* was nothing but infernal, stinking, black Flames and Smoke: and *Saturn* was all melancholy, dreadful, nasty, and dark: for these are the Planets (I don't know why, but all Fortune-tellers hate them) that bring all the Plagues and Mischiefs that we feel upon us, and would exercise their Spite still more, unless they were sometimes mitigated and corrected by the benign and kind Influences of the other Planets. All this and such like Stuff his Genius teaches him. Which he makes give a serious Answer to this idle Question, Whether a Jew or Heathen could be duly and rightly baptized in the Waters of *Venus*? Of him too he learns that the Heaven of the fix'd Stars is not made of solid Matter, but of a thin fluid, [105] wherein an innumerable company of Stars and Suns lie floating here and there, not chain'd down to any Place, (thus far he's in the right) and describing in the Space of a Day these prodigious Circles round the Earth. He forgets here, if there were such a Motion, with what an incredible swiftness they would fly off from every part of their Orbits. But I suppose the Intelligences that he has plac'd in them are to take care of that, those Angels that preside over, and regulate their Motions. And in that he follows a company of Doctors that harbour'd that idle fancy of *Aristotle* upon no Account or Consideration. But *Copernicus* has freed those Intelligences of all that Labour and Trouble, only by bringing in the Motion of the Earth: which, if upon no other Account, every one that is not blind purposely, must own to be necessary upon this. I dare say *Kircher*, if he had dar'd freely to speak his Mind, could have afforded us better sort of Things than these. But when he could not have that liberty, I think he might as well have [106] let the whole Matter alone. But enough of this; let's have done with this famous Author: And now that we have ventur'd to place Spectators in the Planets, let us examine each of them, and see what their Years, Days, and Astronomy are.

The System of the Planets in Mercury. To begin with the innermost and nearest the Sun: We know that *Mercury* is three times nearer that vast Body of Light than we are. Whence it follows that they see him three times bigger, and feel him nine times hotter than we do. Such a degree of Heat would be intolerable to us, and set afire all our dry'd Herbs, our Hay and Straw that we use. And yet there is no doubt but that the Animals there, are made of such a Temper, as to be but moderately warm, and the Plants such as to be able to endure the Heat. The Inhabitants of *Mercury*, it's likely, have the same opinion of us that we have of *Saturn*, that we must be intolerably cold, and have little or no Light, we are so far from the Sun. There's reason to doubt, whether the Inhabitants of *Mercury*, tho' they live so much nearer [107] the Sun, the Fountain of Life and Vigour, are much more airy and

ingenious than we. For if we may guess at them by what we see here, we shall not be obliged to grant it. The Inhabitants of *Africa* and *Brazil*, that have got for their Share the hottest Places in the Earth, being neither so wise nor so industrious as those that belong to colder and more temperate Climates; they have scarce any Arts or Knowledge among them; and those of them that live upon the very Shore, understand little or no Navigation. Nor can I be willing to make all that vast number that must inhabit those two large Planets, *Jupiter* and *Saturn*, and have such noble Attendance, mere dull Blockheads, or without as much Wit as our selves, tho' they are so far more distant from the Sun. The Astronomy of those that live in *Mercury*, and the appearance of the Planets to them, opposite at certain times to the Sun, may be easily conceived by the Scheme of the *Copernican* System in the former Part. At the times of these Oppositions *Venus* and the Earth must needs [108] appear very bright and large to them. For if *Venus* shines so gloriously to us when she is new and horned, she must necessarily in opposition to the Sun, when she is full, be at least six or seven times larger, and a great deal nearer to the Inhabitants of *Mercury*, and afford them Light so strong and bright, that they have no reason to complain of their want of a Moon. What the Length of their Days are, or whether they have different Seasons in the Year, is not yet discovered, because we have not yet been able to observe whether his Axis have any inclination to his Orbit, or what Time he spends in his diurnal Revolution about his own Axis. And yet seeing *Mars*, the Earth, *Jupiter* and *Saturn*, have certainly such Successions, there's no reason to doubt but that he has his Days and Nights as well as they. But his Year is scarce the fourth part so long as ours.

The Inhabitants of *Venus* have much the same Face of Things as those in *Mercury*, only they never see him in opposition to the Sun, which is occasioned [109] by his never removing above 38 degrees, or thereabouts, from it. The Sun appears to them larger by half in his Diameter, and above twice in his Circumference, than to us: and by consequence affords them but twice as much Light and Heat, so that they are nearer our Temperature than *Mercury*. Their Year is compleated in seven and a half of our Months. In the Night our Earth, when 'tis on the other side of the Sun from *Venus*, must needs seem much larger and lighter to *Venus* than she doth ever to us; and then they may easily see, if their Eyes be not weaker than ours, our constant Attendant the Moon. I have often wonder'd that when I have view'd *Venus* when she is nearest to the Earth, and resembled an Half-moon, just beginning to have something like Horns, through a Telescope of 45 or 60 Foot long, she always appeared to me all over equally lucid, that I can't say I observed so much as one Spot in her, tho' in *Jupiter* and *Mars*, which seem much less to us, they are very plainly perceiv'd. [110] For if *Venus* had any such Thing as Sea and Land, the former must necessarily show much more obscure than the other, as anyone may satisfy

himself, that from a very high Mountain will but look down upon our Earth. I thought that perhaps the too brisk Light of *Venus* might be the occasion of this equal appearance; but when I used an Eye-glass that was smok'd for the Purpose, it was still the same Thing. What then, has *Venus* no Sea, or do the Waters there reflect the Light more than ours do, or their Land less? Or rather (which is most probable in my Opinion) is not all that Light we see reflected from an Atmosphere surrounding *Venus*, which being thicker and more solid than that in *Mars* or *Jupiter*, hinders our seeing any thing of the Globe itself, and is at the same time capable of sending back the Rays that it receives from the Sun? For it is certain that if we looked on the Earth from the outside of the Atmosphere, we should not perceive such a difference as we do from a Mountain; but by reason of the interposed [111]Atmosphere, we should observe very little Disparity between Sea and Land. 'Tis the same Thing that hinders us from seeing the Spots in the Moon as plain in the Day as in the Night, because the Vapours that surround the Earth being then enlighten'd by the Rays of the Sun, are an Impediment to our Prospect.

In Mars. But *Mars*, as I said before, has some Parts of him darker than other some. By the constant Returns of which his Nights and Days have been found to be of about the same length with ours. But the Inhabitants have no perceivable Difference between Summer and Winter, the Axis of that Planet having very little or no Inclination to his Orbit, as has been discover'd by the Motion of his Spots. Our Earth must appear to them almost as *Venus* doth to us, and by the Help of a Telescope will be found to have its Wane, Increase, and Full, like the Moon: and never to remove from the Sun above 48 Degrees, by whose Discovery they see it, as well as *Mercury* and *Venus*, sometimes pass over the Sun's Disk. They as seldom see [112]*Venus* as we do *Mercury*. I am apt to believe, that the Land in *Mars* is of a blacker Colour than that of *Jupiter* or the Moon, which is the reason of his appearing of a Copper Colour, and his reflecting a weaker Light than is proportionable to his distance from the Sun. His Body, as I observed before, tho' farther from the Sun, is less than *Venus*. Nor has he any Moon to wait upon him, and in that, as well as *Mercury* and *Venus*, he must be acknowledged inferior to the Earth. His Light and Heat is twice, and sometimes three times less than ours, to which I suppose the Constitution of his Inhabitants is answerable.

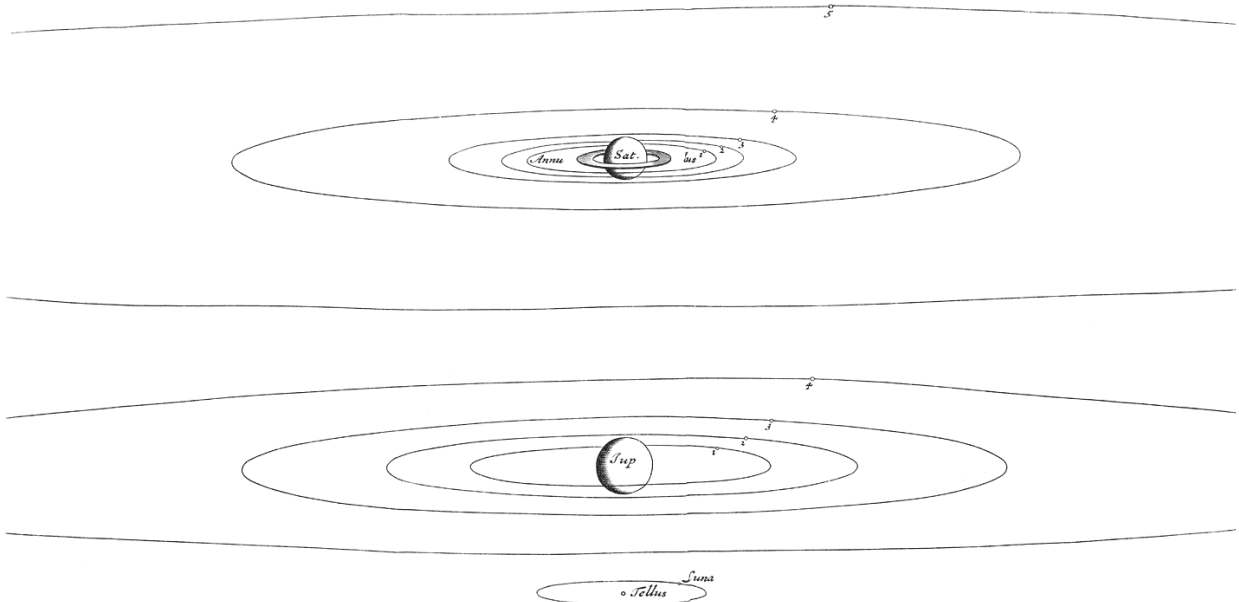


Fig. 3.

p. 113.

Jupiter and Saturn the most eminent of the Planets both for bigness and attendants. If our Earth can claim pre-eminence of the fore-mention'd Planets, for having a Moon to attend upon it, (for its Magnitude can make but a small difference) how much Superiour must *Jupiter* and *Saturn* be to those three and the Earth also? For whether we consider their Bulk, in which they far exceed all the others, or the Number of Moons that wait upon them, it's very probable that they are the chief, the [113]primary Planets in our System, in comparison with which the other four are nothing, and scarce worth mentioning. For the easier Conception of their vast Disparity, I have thought fit to add a Scheme of our Earth, with the Moon's Orbit, and the Globe of the Moon itself, and the Systems of *Fig. 3.* *Jupiter* and *Saturn*, where I have drawn every thing as near the true Proportion as possible. *Jupiter* you see is adorned with four, and *Saturn* with five Moons, all placed in their respective Orbits. The Moons about *Jupiter* we owe to *Galilæo*, 'tis well known: and any one may imagine he was in no small Rapture at the Discovery. The outermost but one, and brightest of *Saturn*'s, it chanc'd to be my lot, with a Telescope not above 12 foot long, to have the first sight of in the Year 1655. The rest we may thank the industrious *Caffini* for, who used the Glaffes of *Jof. Campanus*'s grinding, first of 36, and afterwards of 136 foot long. He has often, and particularly in the Year 1672, shew'd me the Third and Fifth. The First and Second he gave [114]me notice of by Letters in the Year 1684; but they are scarce ever to be seen, and I can't positively say, I had ever that Happiness; but am as satisfied that they are there, as if I had; not in the least suspecting the Credit of that worthy Man. Nay, I am afraid there are One or Two more still behind, and not without

reason. For between the Fourth and Fifth there's a Distance not at all proportionable to that between all the others: Here, for ought I know, there may be a Sixth; or perhaps there may be another without the Fifth that may yet have escaped us: for we can never see the Fifth but in that part of his Orbit, which is towards the West: for which we shall give you a very good reason.

Perhaps when *Saturn* comes into the Northern Signs, and is at a good height from the Horizon (for at the writing of this he is at his lowest) you may happen to make some new Discoveries, good Brother, if you would but make use of your two Telescopes of 170 and 210 Foot long; the longest, and the best I believe now [115] in the World. For tho' we have not yet had an opportunity of observing the Heavens with them (as well by reason of their Unwieldiness, as for the Interruption of our Studies by your Absence) yet I am satisfied of their Goodness by our trial of them one Night, in reading a Letter at a vast distance by the Help of a Light. I cannot but think of those times with Pleasure, and of our diverting Labour in polishing and preparing such Glasses, in inventing new Methods and Engines, and always pushing forward to still greater and greater Things. But to return to the Figures, of which there remains something further to be said.

The proportion of the Diameter of Jupiter, and of the Orbs of his Satellites, to the Orbit of the Moon round the Earth. I have there made the Diameter of *Jupiter* about two third parts of our distance from the Moon: for the Diameter of *Jupiter* is above twenty times bigger than that of the Earth; which is about a thirtieth part of the Moon's distance. The Orbit of the outermost of *Jupiter's* Satellites is to that of the Moon round the Earth, as 8 and $\frac{1}{2}$ is to 1. And each of these Moons, by the Shadow they make upon *Jupiter*, [116] cannot be less than our Earth. Their *The Periods of Jupiter's Moons*. Periods, that I may not omit them, are according to *Caffini's* Account these. That of the innermost is one day, 18 hours, 28 minutes, and 36 seconds. The Second spends 3 days, 13 hours, 13 min., 52 seconds in going round him. The Third 7 days, 3 hours, 59 min., 40 sec. The Fourth 16 days, 18 hours, 5 min., 6 sec. The Distance of the innermost from *Jupiter* himself is $2\frac{5}{6}$ of his Diameters. That of the Second is 4 and a half: Of the Third 7 and one sixth part: Of the Fourth 12 and two thirds, of the same *And Saturn's*. Diameters. The Innermost of *Saturn's* Satellites moves round him in 1 day, 21 hours, 18 min., 31 sec. The Second in 2 days, 17 hours, 41 min., 27 sec. The Third in 4 days, 13 hours, 47 min., 16 sec. The Fourth in 15 days, 22 hours, 41 min., 11 sec. The Fifth in 79 days, 7 hours, 53 min., 57 sec. Their Distances from the Center of *Saturn* are, that of the first almost one, that is 39 fortieth parts of the Diameter of his Ring; that of the second one and a quarter of those Diameters; [117] of the third one and three quarters of them; of the fourth four, or according to my Calculation, but 3 and a half; of the 5th 12, which were found with vast Pains and Labour.

Now can any one look upon, and compare these Systems together, without being amazed at the vast Magnitude and noble Attendance of these two Planets, in respect of this little Earth of ours? Or can they force themselves to think, that the wise Creator has disposed of all his Animals and Plants here, has furnish'd and adorn'd this Spot only, and has left all those Worlds bare and destitute of Inhabitants, who might adore and worship him; or that all those prodigious Bodies were made only to twinkle to, and be studied by some few perhaps of us poor Mortals?

This proportion true according to all modern Observations. I do not doubt but there will be some who will think we are very much mistaken about the Magnitude of these Planets. For will you pretend to make them who are taken up in admiring the Largeness of this Globe, [118] its multitude of Nations, Cities, and Empires; can you pretend I say to make them ever believe that there are Places in comparison of which the Earth is as inconsiderable as this Figure would make it? But they ought to be inform'd, that these Proportions are those which the best Astronomers of this Age have agreed upon. For if the Earth be distant from the Sun ten or eleven thousand of its own Diameters, according to the Accounts of Monsieur *Cassini* in *France*, and Mr. *Flamsted* in *England*, wherein they made use of very exact Observations of the Parallaxes of *Mars*; or if, according to a very probable Conjecture of mine, it be distant twelve thousand, then the Magnitudes of the other Orbs will very near answer the Proportions here settled.

The apparent magnitude of the Sun in Jupiter, and a way of finding what Light they there enjoy. But to return to *Jupiter*. The Sun appears to them who are upon it five times less than to us, and consequently they have but the five and twentieth part of the Light and Heat that we receive from it. But that Light is not so weak as we imagine, as is [119] plain by the Brightness of that Planet in the Night; and also from hence, that when the Sun is so far eclipsed to us, as that only the 25th part of his Disk remains uncovered, he is not sensibly darken'd. But if you have a mind exactly to know the Quantity of Light that *Jupiter* enjoys, you may take a Tube of what Length you please. Let one end of it be closed with a Plate of Brass, or any such thing, in the middle of which there must be a Hole, whose Breadth must have the same proportion to the length of the Tube, as the Chord of 6 Minutes bears to the Radius; that is, about as one is to 570. Let the Tube be turned so to the Sun, that no Light may fall upon a white Paper placed at the End of it, but what comes through the little Hole at the other end of the Tube. The Rays that come through this will represent the Sun upon the Paper of the same Brightness that the Inhabitants of *Jupiter* see it in a clear Day. And if removing the Paper you place your Eye in the same Place, you will see the Sun of the same Magnitude and [120] Brightness as you would were you in *Jupiter*.

And in Saturn. If you make the Hole twice as little in breadth, you will see the same in *Saturn*. And altho' his Light be but the hundredth part of ours, yet you see it makes him shine tolerably bright in a dark Night. But in both these Planets, if there ever be any cloudy Days, it must be very dark in comparison of us; yet without doubt the Inhabitants have no more reason to complain of the want of Light, than our Owls and Bats, to whom the Twilight or the Night itself is more agreeable than the Brightness of the Day.

In Jupiter their days are five Hours. But it's a little strange, that when *Jupiter* is so much bigger than our Planet, their Days and Nights should be but five of our Hours. By this we may see that Nature has not observ'd that proportion that their Bulk seems to require, seeing in *Mars* the Days are very little different from ours. But in the length of their Years, that is, in the Revolution of the Planets round the Sun, there is an exact proportion to ^[121]their distances from the Sun followed. For as the Cubes of their Distances, so are the Squares of their Revolutions, as *Kepler* first found out. Which proportion the Moons of *Jupiter* and *Saturn* keep in their Courses round those *Always of the same length.* Planets. As the Years and Days in *Jupiter* are different from ours in this respect, so are the Days in another; namely, that they are all of the same length. For they there enjoy a perpetual Equinox, their Axis having little or no inclination to their Orbit, as the Earth's has, as has been discovered by Telescopes. The Countries that lie near their Poles have little or no Heat, by reason the Rays of the Sun fall so obliquely upon them; but then they are freed from the Inconveniency that ours are troubled with, of tedious long half-year Nights, and have the constant returns of Day and Night every five Hours. Indeed such short Days would not be agreeable to us, but we think our selves much better done by, that ours are more than twice as long, tho' upon no other account, but that whatever is our own, we are apt to imagine, must be best. ^[122]

The rest of the Planets are so near the Sun (*Mars* himself never being above 18 degrees from it) that in *Jupiter* they have the light only of *Saturn*. But we cannot deny but that their four Moons stand them in greater stead than our one doth us, if 'twere only that they seldom know any such Thing as to be without Moonshiny Nights. And they are of great Advantage to them, as we said before, in their Navigation, if they have any such thing. Not to mention the pleasant Sights of their frequent Conjunctions and Eclipses, Things that they are seldom a Day without.

Saturn enjoys all those Pleasures and Advantages in a still higher Degree, as well for his five Moons, as for the delightful Prospect that the Ring about him affords his Inhabitants Night and Day. But we will give an account of their Astronomy, as we have done of the rest of the Planets.

They see the fix'd Stars just as we do. And first of all we shall observe what we might have remark'd before, but which will be more strange here, that the fix'd Stars appear to them of [123]the same Figure and Magnitude, and with the same degree of Light that they do to us: and this, by reason of their immense distance, of which we shall have occasion to speak by and by. In comparison with which the Space that a Bullet-shot out of a Cannon could travel in 25 Years, would be almost nothing.

Their Astronomers have all the same Signs of the Bear, the Lion, Orion, and the rest, but not turning upon the same Axis with us: for that's different in all the Planets.

As *Jupiter* can see no Planet but *Saturn*, so *Saturn* knows of no Planet but *Jupiter*; which appears to him much as *Venus* doth to us, never removing above 37 Degrees from the Sun. The Length of their Days I cannot determine: But if from the Distance and Period of his innermost Attendant, and comparing it with the innermost of *Jupiter*'s, a Man may venture to give a Guess, they are very little different from *Jupiter*'s, 10 Hours or somewhat less. But whereas in *Jupiter* these are equally divided between [124]Light and Darkness, the Inhabitants of *Saturn* must perceive a more sensible difference than we, especially between Summer and Winter. For our Axis inclines to the Plane of the Ecliptick but 23 degrees and a half, but there's above 31. Upon this Account his Moons must decline very much from the Path that the Sun seems to move in, and his Inhabitants can never have a full Moon but just at the Equinoxes; Two of which fall out in 30 of our Years. 'Tis this Position of the Axis too that is the Cause of those delightful Appearances, and wonderful Prospects that its Inhabitants enjoy: For the better understanding of which I shall draw a Figure of *Saturn* with his Ring about him: in which the Proportion between the Diameters of the Globe and Ring is as 9 to 4. And the empty Space between them is of the same Breadth with the Ring itself. All Observations conspire to prove that That is of no great Thickness, altho' if we should allow it six hundred *German* Miles, I think, considering its Diameter, we should not overdo the Matter.

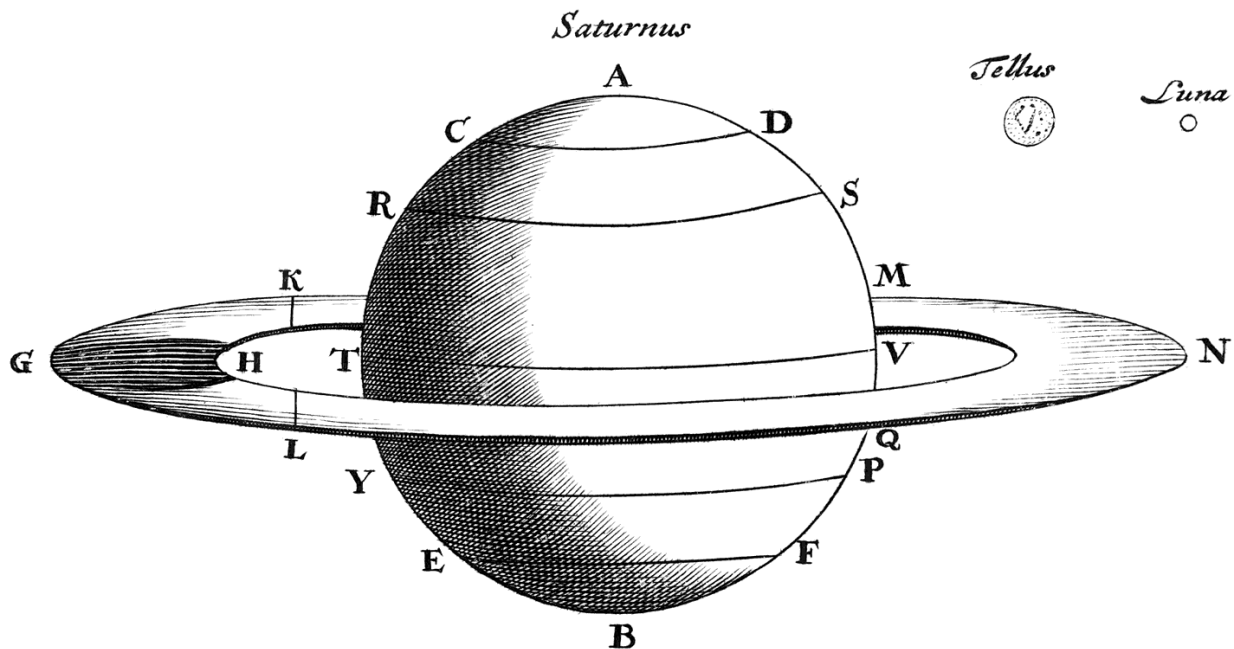


Fig. 4.

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Suppose then, agreeable to what has been said, the Globe of *Saturn*, *Fig. 4.* whose Poles are A, B. G N is the Diameter of the Ring, as you view it sideways, representing a narrow Oval. Those that live about the Poles within the Arches C A D, E B F, each of which are 54 Degrees, (if the Cold will suffer any Body to live there) never have a Sight of the Ring. *The Appearances of the Ring in Saturn.* From all other parts it is continually to be seen for fourteen Years and nine Months, which is juft half their Year. The other Half it is hid from their View. Those then that dwell between the Polar Circle C D, and the Equator T V, all that time that the Sun enlightens the Part opposite to them, have every Night the Sight of a Piece of it H G L, much in the Shape of a fhining Bow, which comes from the Horizon, but is darken'd in the Middle by the Shadow of *Saturn* G H, which reaches moft commonly to the outermoft Rim of it. But after Midnight that Shadow by little and little begins to move towards the right Hand to those in the Northern, but the Left [126] to those in the Southern Hemisphere. In the Morning it difappears, leaving behind it a Likeness indeed of a Bow, but much paler and weaker than our Moon is in the Day time. For they, as I said before, have an Atmosphere, or an Air furrounding them enlightened by the Sun. Otherwise Night and Day they would have their Ring, their Moons, and all the fix'd Stars, equally conspicuous. Another thing that must make the Sight of their Ring very curious, is, that by some Spots in it, it is discover'd to turn round upon it self: A thing that those that are so near cannot but take notice of, when we that live at this

Distance can defcry a great Inequality, the infide of it being brighter much than the outfide is. When the Shadow of the Globe falls upon that part of the Ring G H, the Shadow of the Ring at the fame time darkens another Part of the Globe about P F, which otherwife would have the Sun upon it. So that there is always a Zone of the Globe P Y F E, fometimes of a larger extent than at others, which is depriv'd of the Sight [127] both of the Sun and Ring for a confiderable time, the latter of which hides fome part of the Stars from it too. And certainly an amazing Thing it muft be, all of a fudden to have the Sun intercepted and to become as dark as Midnight, without feeing any Caufe of fuch an Accident. All which time their Moons are their only Comfort. The other half of the Year the Hemisphere T B V enjoys the fame Light that T A U before did, and then this undergoes thofe long Eclipses that That before fuffer'd. At the Equinoxes, when the Sun is in the fame Plane with the Ring, the Inhabitants of *Saturn* cannot well perceive it: no not even we with our Glaffes, by reafon of its Darknefs. This happens when *Saturn*, view'd from the Sun, is advanced one and twenty degrees and a half in *Virgo* or *Pifces*, as I have fhew'd formerly in my Syftem of *Saturn*: Where there is an Account given of the Rifings of the Sun above the Ring, throughout all the Saturnian Year. [128]

With *Saturn* in this Scheme you have the Globes of the Earth and Moon drawn in their true proportion, to put you in mind again of a Thing worth remembring, viz. how very fmall our Habitation is when compar'd with that Globe or the Ring about it. And now any one, I fuppofe, can frame to himfelf a Picture of the Night in *Saturn*, with two Arches of the Ring, and five Moons fhining about, and adorning him. This then is what I had to fay to the primary Planets.

We are now come a little lower, to make an enquiry into the Attendants of thefe Planets, efpecially our own. And here we fhall not only confider their Aftronomy, but fhall alfo fearch into their Furniture and Ornament, if they are found to have any fuch thing, which we have deferred confidering till now.

Very little to be faid of the Moon. And here one would think that when the Moon is fo near us, and by the Means of a Telescope may be fo nicely and exactly obferv'd, it fhould afford us Matter for more probable [129] Conjectures than any of the other remote Planets. But it is quite otherwife, and I can fcarce find any thing to fay of it, becaufe I have not a Planet of the fame Nature before my Eyes, as in all the primary ones I have. For they are of the fame kind with our Earth; and feeing all the Actions, and every thing that is here, we may make a reafonable Conjecture at what we cannot fee in thofe Worlds.

The Guards of Jupiter and Saturn of the fame nature with our Moon. But this we may venture to fay, without fear, that all the Attendants of *Jupiter* and *Saturn* are of the fame Nature with our

Moon, as going round them, and being carried with them round the Sun juft as the Moon is with the Earth. Their Likeneſs reaches to other Things too, as you'll ſee by and by. Therefore whatſoever we can with reaſon affirm or conjecture of our Moon (and we may ſay a little of it) muſt be ſuppos'd with very little Alteration to belong to the Satellites of *Jupiter* and *Saturn*, as having no reaſon to be at all inferior to that.

The Moon hath Mountains. The Surface of the Moon then is found, by the leaſt Telescopes of about [130]three or four Foot, to be diversifi'd with long Tracts of Mountains, and again with broad Valleys. For in thoſe Parts oppoſite to the Sun you may ſee the Shadows of the Mountains, and often diſcover the little round Valleys between them, with a Hillock or two perhaps riſing out of them. *Kepler* from the exact roundneſs of them would prove that they are ſome vaſt work of the rational Inhabitants. But I can't be of his mind, both for their incredible Largeneſs, and that they might eaſily be occaſioned by natural Cauſes. Nor can I find any thing like Sea there, tho' he and many others are of the contrary Opinion I know. For thoſe vaſt Countries which appear darker than the other, commonly taken for and called by the Names of Seas, are diſcover'd with a good long Telescope, to be full of little round Cavities; whoſe Shadow falling within themſelves, makes them appear of that Colour: and thoſe large Champains there in the Moon you will find not to be always even and ſmooth, if you look carefully [131]upon them: neither of which two *But no Sea.* Things can agree to the Sea. Therefore thoſe Plains in her that ſeem brighter than the other Parts, muſt conſiſt, I ſuppoſe, of a whiter ſort of Matter than they. Nor do I believe *Nor Rivers.* that there are any Rivers, for if there were, they could never eſcape our Sight, eſpecially if they run between the Hills as ours do. Nor have they *Nor Clouds.* any Clouds to furniſh the Rivers with Water: For if they had, we ſhould ſometimes ſee one part of the Moon darkened by them, and ſometimes another, whereas we have always the ſame Proſpect of her.

Nor Air, and Water. 'Tis certain moreover, that the Moon has no Air or Atmoſphere ſurrounding it as we have. For then we could never ſee the very outermoſt Rim of the Moon ſo exactly as we do, when any Star goes under it, but its Light would terminate in a gradual faint Shade, and there would be a ſort of a Down as it were about it; not to mention that the Vapours of our Atmoſphere conſiſt of Water, and conſequently that where there are no Seas [132]or Rivers, there can be no Atmoſphere. This is that notable difference between the Moon and us that hinders all probable Conjectures about it. If we could but once be ſure that there were Seas and Rivers in it, it would be no weak Argument to prove that it has alſo all other Furniture which belongs to our Earth, and the Opinion of *Xenophanes* might be true, that it has its Inhabitants, Cities, and Mountains. But as 'tis, I cannot imagine how any Plants or Animals, whole whole nourishment comes from liquid Bodies, can thrive in a dry, waterleſs, parch'd Soil.

The Conjecture of its Plants and Animals very dubious. What then, is it credible that this great Ball was made for nothing but to give us a little Light in the Night-time, or to raise our Tides in the Sea? May there not be some People there that may have the Pleasure of seeing our Earth turn upon itself, presenting them sometimes with a Prospect of *Europe* and *Africa*, and then of *Asia* and *America*; sometimes half of it bright, and sometimes full? And must all those Moons round *Jupiter* and *Saturn* [133] be condemned to the same Usefulness? I do not know what to say concerning it, because I know of nothing like them to found a Conjecture upon. And yet 'tis not improbable that those great and noble Bodies have somewhat or other growing and living upon them, though very different from what we see and enjoy here. Perhaps their Plants and Animals may have another sort of Nourishment there. Perhaps the Moisture of the Earth there is but just sufficient to cause a Mist or Dew, which may be very suitable to the Growth of their Herbs. This I remember is *Plutarch's* Opinion, in his Dialogue upon this Subject. For in our Earth a very little Water drawn from the Sea into Dew, and falling down again upon the Herbs, would be sufficient for all our Needs, without any Rain or Showers. But these are mere Guesses, or rather Doubts, but yet they are the *Jupiter's and Saturn's Moons turn always the same Side to them.* best we can make of this, and all those other Moons: for, as I said before, they are all of the same nature, which is proved likewise by this, that as our [134] Moon can afford us the Sight never but of one Side of her, so they turn always the same Face to their primary Planets. It may perhaps seem strange, how we should come to know this; but 'tis no hard matter, after that Observation which I just now made, that the outermost of *Saturn's* Moons can never be seen but when she is on the West-side of her Planet. The reason of which is plainly this, that one Side of her is darker, and does not reflect the Light so much as the other, which when it is turned towards us, we cannot see by reason of its weak Light. This always happening when 'tis East of him, and never on the other Side, is a manifest proof that she always keeps the same Side toward *Saturn*. Now since the outermost of *Saturn's* and our Moon carry themselves thus to the Planets round which they move, who can well doubt it of all the rest round *Jupiter* and *Saturn*? And there's a very good reason for it, namely, that the matter of which those Moons consist, being heavier, and more solid on the Side that isaverse from us, than on [135] that which we have the Sight of, does consequently fly with a greater force from the Centre of its Orbit: for otherwise, according to the Laws of Motion, it should turn the same Side always, not to its Planets, but to the same fix'd Stars.

This Position of the Moons, in respect of their Planets, must occasion a great many very surprizing Appearances to their Inhabitants, if they have any, which is very doubtful, but may for the present be suppos'd. An enquiry into our Moon may serve for all the rest. Its Globe is divided into two Parts, in such a manner, that those who live on one Side never

lose the light of us, and those on the other never enjoy it. Except only some few who live on the Confines of each of these, who lose us, and see us again by turns. The Earth to them must seem *The Astronomy of the Inhabitants of the Moon*. much larger than the Moon doth to us, as being in Diameter above four times bigger. But that which is most surprising, is, that Night and Day they see it always in the very same part of the Heaven, as if it never moved: some of them as if 'twas falling [136] upon their Heads: others somewhat above the Horizon, and others always in the Horizon, still turning upon it self, and presenting them every twenty four Hours with a View of all its Countries, even of those that lie near the Poles (I could wish my self in the Moon only for the sight of them) yet unknown and undiscovered by us. They have it in its monthly Wane and Increase, they see it half, and horned, and full, by turns, just as we do the Body of the Moon. But the Light that they receive of us is five times larger than what we receive from them. So that in dark Nights that part that hath the Advantage of being towards us, receives a very glorious Light from us, tho' *Kepler* thought otherwise. Their Days are always of the same Length with their Nights; and the Sun rising and setting to them but once in one of our Months, makes the time both of their Light and Darknests to be equal to 15 of our Days. If their Bodies were of the same Materials with ours, those that have the Sun pretty high in their Horizon, would be almost roasted [137] in such long Days. For the Sun is not farther from them than he is from us. This will be the Case of those that live upon the Borders of the two Hemispheres we mentioned; but those that live under the Poles of the Moon will be just about as hot as our Whale-fishers about *Ifland* and *Nova Zemla* are, in the Summer-time: who are in so little danger of being roasted, that in the middle of their Summer, in their Days of three Months length, they very often find it extreme Cold. I call those the Poles of the Moon, round which the fix'd Stars seem to turn to its Inhabitants, which are different from ours, and also from those of the Ecliptick, although they move round these latter, at the distance of five Degrees, in a period of nineteen Years. Their Year they count by the Motion of the Stars, and their return to the Sun, and 'tis the same with ours. They can easily do it, because they have the Stars Day and Night, notwithstanding the Light of the Sun: for they have no Atmosphere (which is the only reason that we don't every [138] Day enjoy the same Sight) to hinder their Observations. Nor have they any Clouds to obstruct their View, so that it is easier for them to find out the Courses of the Planets, but more difficult to make a true System of them. For they will be apt to lay a wrong Foundation, by supposing that their Earth stands still, which will lead them into more dangerous Errors than *This may be applied to the Moons about Jupiter and Saturn*. ever it did us. All that I have said belongs as well to *Jupiter's* and *Saturn's* Satellites as to our Moon, in respect of the Planets they move round. The Length of their Day and Night is always equal to the Time of their Revolution: For example, the fifth Moon moves round *Saturn* in 80 Days, and the Days and Nights there are equal to Forty of ours. Both their Summer

and Winter (*Saturn* moving round the Sun in thirty Years) are fifteen Years long. Therefore it is impossible but that their way of living must be very different from ours, having such tedious Winters, and such long watching and sleeping times.

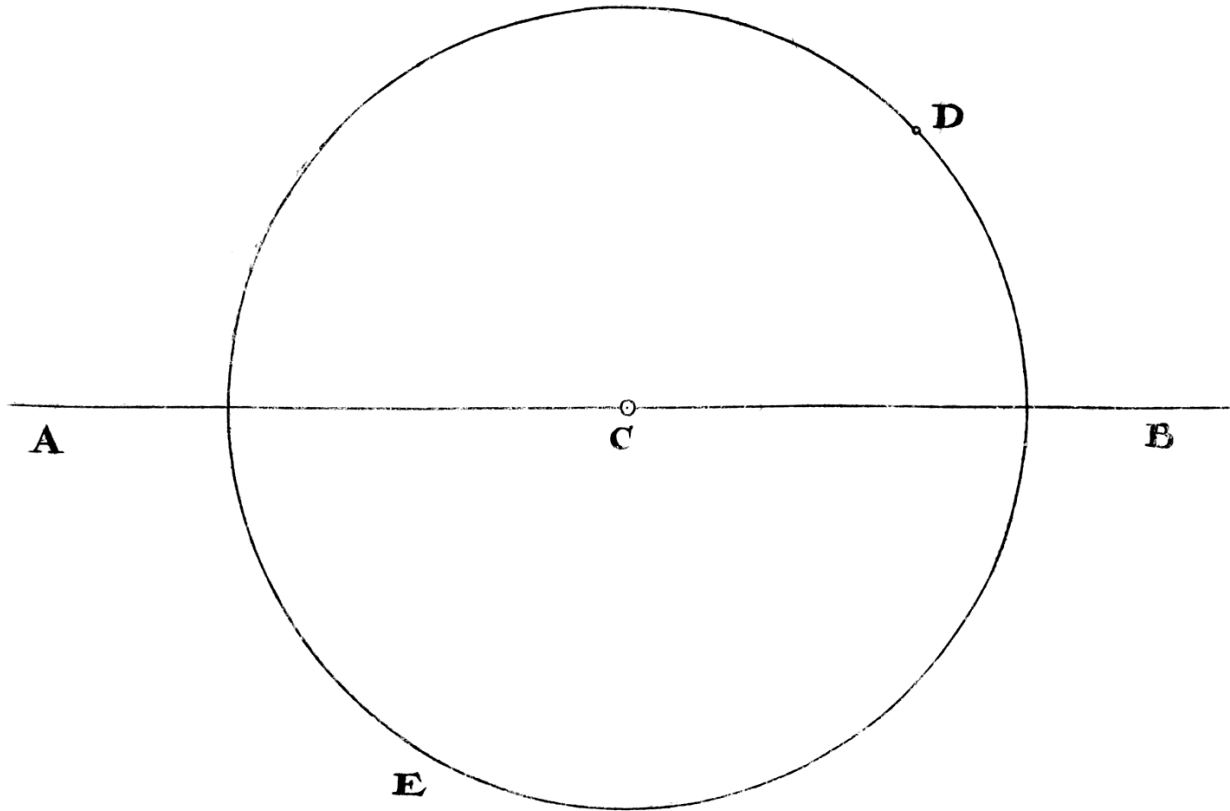


Fig. 5.

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Having thus explained the primary and secondary Planets round the Sun, we should next set about the third Sort, the Sun and fix'd Stars; but before we do that, it would be worth while to set before you at once, in a clearer and more plain Method than hitherto, the Magnificence and Fabrick of the Solar System. Which we can't possibly do in so small a Space as one of our Leaves will but admit of, because the Bodies of the Planets are so prodigiously small in companion of their Orbs. But what is wanting in Figure shall be made up in Words. Going back then to the first Scheme, suppose another *Fig. 1.* like it, and proportionable, drawn upon a very large smooth Plain; whose outermost Circle representing the Orb of *Saturn*, must be conceived three hundred and sixty Foot in Semidiameter. In which you must place the Globe and Ring of *Saturn* of that *Fig. 2.* Bigness as the 2d Figure shows you. Let all the other Planets be supposed every one in his

own Orbit, and in the middle of all the Sun, of the same Bigness that That Figure represents, [140]namely, about four Inches in Diameter. And then the Orbit or Circle in which the Earth moves, which the Astronomers call the *Magnus Orbis*, must have about six and thirty Foot in Semidiameter. In which the Earth must be conceived moving, not bigger than a grain of Millet, and her Companion the Moon scarcely perceivable, moving round her in a Circle a little more than two Inches Diameter, as in *Fig. 5.* the Figure here adjoined, where the Line A B represents a small portion of that Circle which the Earth moves in: the small Circle therein C is the Earth, and the Circle D E the Path of the Moon round it, in which the Body of the Moon is D.

The outermost of *Saturn's* Moons moves in an Orbit whose Semidiameter is 29 Inches; that of *Jupiter* in a somewhat smaller, whose Semidiameter is 19 and a quarter.

And thus we have a true and exact Description of the Sun's Palace, where the Earth will be Twelve thousand of its Semidiameters distant from him, which in *German Miles* makes above [141]seventeen Millions. But perhaps we may have a clearer Comprehension of this vast Length, by comparing it with some very swift Motion after the Example of *Hesiod* the Poet, who imagin'd that an Anvil let fall from the Top of Heaven, reach'd the Earth the tenth Day of its Journey, and in ten more arriv'd at the Bottom of Hell, the end of it: so making the Earth the mid-way between Heaven and Hell. I shan't make use of the Anvil, but of something as good, namely, a Bullet shot out of a great Gun, which may travel perhaps in a Moment, or Pulse of an Artery, about a hundred Fathom, as is proved by those Experiments that *Mersennus* in a Treatise of his relates; by which the Sound was found to extend itself eighty hundredth parts in the *The immense distance between the Sun and Planets illustrated.* same time. I say then, that supposing a Bullet to move with this Swiftness from the Earth to the Sun, it would spend 25 Years in its Passage. To make a Journey from *Jupiter* to the Sun, would require 125, and from *Saturn* thither 250 Years. This account depends upon the measure of the Earth's [142]Diameter, which, according to the accurate Observations of the *French*, is 6538594 times six *Paris Feet*, one Degree being 57060 of that Measure. This shows us how vast those Orbs must be, and how inconsiderable this Earth, the Theatre upon which all our mighty Designs, all our Navigations, and all our Wars are transacted, is when compared to them. A very fit Consideration, and Matter of Reflection, for those Kings and Princes who sacrifice the Lives of so many People, only to flatter their Ambition in being Masters of some pitiful Corner of this small Spot. But to return to the matter in hand, now we have given you an account of the Sun's proportion to those Orbs and Bodies, we'll see what more we can say of him.

No ground for Conjecture in the Sun. And some have thought it not improbable but that the Sun himself has also his Inhabitants. But upon what reason I cannot imagine, there being less

ground for a Probability in him than in the Moon. For we are not yet sure, whether he be a solid or liquid Globe; altho', if my Notion of Light [143] be true, upon that account I should rather think him liquid: which his Roundness and equal distribution of his Light to all parts are an Argument for. For that very small inequality on his Surface, which is discovered by the Telescopes, (and that not always neither) which makes Men fancy they see boiling Seas and belching Mountains of Fire, is nothing but the trembling Motion of the Vapours our Atmosphere is full of near the Earth; which is likewise *The Faculae in the Sun not easily seen.* the Cause of the Stars twinkling. Nor could I ever have the Luck to discern those bright Spots in the Sun which they boast as much of as they do of his dark ones, which latter I have very often seen; so that I have very good Reason to doubt whether there be any thing in the Sun brighter than the Sun itself. For by the most exact Observations, I could never find any such pretended to be seen any where but just about his dark Spots; and it is no great wonder that those Parts which are so near the darker, should appear somewhat *By reason of its Heat no Inhabitants like ours can live in the Sun.* brighter than the rest. That the Sun is extremely hot and fiery, is beyond [144] all dispute, and such Bodies as ours could not live one Moment in such a Furnace. We must suppose a new sort of Animals then, such as we have no Idea or Likeness of among us, such as we can neither imagine nor conceive: which is as much as to say, that we can make no Supposition at all about them. No doubt that glorious and vast Body was made for some noble End and Use, and fram'd with excellent Design. And I think we all very well know and feel its Usefulness in that effusion of Light and Heat to all the Planets round it; in the Preservation and Happiness of all living Creatures, and that not only in our Ball, but in those vast Globes of *Jupiter* and *Saturn*, not contemptible when compared with its own. These are such great, such wise Ends, that it is not strange that the Sun should have been made, if it had been only upon their account. For, as for *Kepler's* Fancy, that he hath another Office, namely, to help on the Motion of the Planets in their own Orbs, by turning about his own Axis (which he would fain establish in his Epitome of the *Copernican* [145] System) I shall give good Reasons why I cannot assent to it.

The fix'd Stars so many Suns. Before the Invention of Telescopes, it seemed to contradict *Copernicus's* Opinion, to make the Sun one of the fix'd Stars. For the Stars of the first Magnitude being esteem'd to be about three Minutes Diameter; and *Copernicus* (observing that tho' the Earth changed its Place, they always kept the same distance from us) having ventur'd to say that the *Magnus Orbis* was but a Point in respect of the Sphere in which they were placed, it was a plain Consequence that every one of them that appeared any thing bright, must be larger than the Path or Orbit of the Earth: which is very absurd. This is the principal Argument that *Tycho Brahe* set up against *Copernicus*. But when the Telescopes took away those Rays of the Stars which appear when we look

upon them with our naked Eye, (which they do best when the Eye-glass is black'd with Smoke) they seem'd just like little shining Points, and then that Difficulty vanish'd, and the Stars may yet be so many Suns. Which is [146]the more probable, because their Light is certainly their own: for it's impossible that ever the Sun should send, or they reflect it at such a vast Distance. This is the Opinion that commonly goes along with *Copernicus's* System. *They are not all in the same Sphere.* And the Patrons of it do also with reason suppose, that all these Stars are not in the same Sphere, as well because there's no Argument for it, as that the Sun, which is one of them, cannot be brought to this Rule. But it's more likely they are scatter'd and dispers'd all over the immense Spaces of the Heaven, and are as far distant perhaps from one another, as the nearest of them are from the Sun.

Here again too I know *Kepler* is of another Opinion in his Epitome of *Copernicus's* System, that we mention'd above. For tho' he agrees with us, that the Stars are diffus'd through all the vast Expanse of the Heavens, yet he cannot allow that they have as large an empty Space about them as our Sun has. For then 'twas his Opinion, we should see but very few, and those of very different Magnitudes: *For, seeing [147]the largest of all appear so small to us, that we can scarce observe or measure them with our best Instruments; how must those appear that are three or four times farther from us? Why, supposing them no larger than these, they must seem three or four times less, and so on 'till a little farther they will not be to be seen at all: Thus we shall have the sight of but very few Stars, and those very different one from another;* Whereas we have above a Thousand, and those not considerably bigger or less than one another. But this by no means proves what he would have it; and his Mistake was chiefly, that he did not consider the Nature of Fire and Flame which may be seen at such distances, and at such small Angles as all other Bodies would totally disappear under. A thing that we need go no farther than the Lamps set along the Streets to prove. For altho' they are a hundred Foot from one another, yet you may count Twenty of them in a continued Row with your Eyes, and yet the twentieth Part of them scarce makes an Angle of six Seconds. Certainly [148]then the glorious Light of the Stars must do much more than this; so that it's no wonder we should see a Thousand or two of them with our bare Eyes, and with a Telescope discover twenty times that number. But *Kepler* had a private Design in making the Sun thus superiour to all the other Stars, and planting it in the Middle of the World, attended with the Planets: For his Aim was hereby to strengthen his Cosmographical Mystery, that the Distances of the Planets from the Sun are in a certain proportion to the Diameters of the Spheres that are inscribed within, and circumscribed about *Euclid's* Regular Bodies. Which could never be so much as probable, except there were but one Chorus of Planets moving round the Sun, and so the Sun were the only one of his kind.

But that whole Myſtery is nothing but an idle Dream taken from *Pythagoras* or *Plato's* Philoſophy. And the Author himſelf acknowledges that the Proportions do not agree ſo well as they ſhould, and is fain to invent two [149]or three very filly Excuses for it. And he uſes yet poorer Arguments to prove that the Univerſe is of a ſpherical Figure, and that the Number of the Stars muſt neceſſarily be finite, becauſe the Magnitude of each of them is ſo. But what is worſt of all is, that he ſettles the Space between the Sun and the Concavity of the Sphere of the fix'd Stars, to be fix hundred thouſand of the Earth's Diameters. For this reaſon, which he has no Foundation for, that as the Diameter of the Sun is to that of the Orbit of *Saturn*, which he makes to be as 1 to 2000, ſo is this Diameter to that of the Sphere of the fixed Stars. I cannot but wonder how ſuch things as theſe could fall from ſo ingenious a Man, and ſo great an Aſtronomer. But I muſt be of the ſame Opinion with all the greateſt Philoſophers of our Age, that the Sun is of the ſame Nature with the fix'd Stars. And this will give us a greater Idea of the World, than all thoſe other Opinions. *The Stars have Planets about them like our Sun.* For then why may not every one of theſe Stars or Suns have as great a Retinue as our Sun, of Planets, with their [150]Moons, to wait upon them? Nay, there's a manifeſt reaſon why they ſhould. For if we imagine our ſelves placed at an equal diſtance from the Sun and fix'd Stars; we ſhould then perceive no difference between them. For, as for all the Planets that we now ſee attend the Sun, we ſhould not have the leaſt glimpe of them, either becauſe their Light would be too weak to affect us, or that all the Orbs in which they move would make up one lucid Point with the Sun. In this Station we ſhould have no occaſion to imagine any difference between the Stars, and ſhould make no doubt if we had but the Sight, and knew the Nature of one of them, to make that the Standard of all the reſt. We are then plac'd near one of them, namely, our Sun, and ſo near as to diſcover fix other Globes moving round him, ſome of them having others performing them the ſame Office. Why then may not we make uſe of the ſame Judgment that we would in that caſe; and conclude, that our Star has no better attendance than the others? So that [151]what we allowed the Planets, upon the account of our enjoying it, we muſt likewise grant to all thoſe Planets that ſurround that prodigious number of Suns. They muſt have their Plants and Animals, nay and their rational Creatures too, and thoſe as great Admirers, and as diligent Obſervers of the Heavens as our ſelves; and muſt conſequently enjoy whatſoever is ſubſervient to, and requiſite for ſuch Knowledge.

What a wonderful and amazing Scheme have we here of the magnificent Vaſtneſs of the Univerſe! So many Suns, ſo many Earths, and every one of them ſtock'd with ſo many Herbs, Trees, and Animals, and adorn'd with ſo many Seas and Mountains! And how muſt our Wonder and Admiration be increaſed when we conſider the prodigious Diſtance and Multitude of the Stars?

That their Distance is so immense, that the Space between the Earth and Sun (which is no less than Twelve thousand of the Earth's Diameters) is almost nothing when compar'd to it, has more Proofs than one to confirm [152]it. And this among the rest. If you observe two Stars near one another, as for example those in the middle of the Great Bears Tail, differing very much from one another in Clearness, notwithstanding our changing our Position in our Annual Orbit round the Sun, and that there would be a Parallax were the Star which is brighter nearer to us than the other, as is very probable it is, yet whatever Part of the Year you look upon them, they will not in the least have altered their distance. Those that have hitherto undertook to calculate their Distance, have not been able perfectly to compass their Design, by reason of the extreme Niceness and almost Impossibility of the Observations requisite for their Purpose. The only Method that I see remaining, to come at any tolerable Probability in so difficult a Case, I shall here make use of. Seeing then that the Stars, as I said before, are so many Suns, if we do but suppose one of them equal to ours, it will follow that its distance from us is as much greater than that of the Sun, as its apparent [153]Diameter is less than the Diameter of the Sun. But the Stars, even those of the first Magnitude, though view'd through a Telescope, are so very small, that they seem only like so many shining Points, without any perceivable Breadth. So that such Observations can here do us no good. When *A way of making a probable guess at the distance of the Stars.* I saw this would not succeed, I studied by what way I could so lessen the Diameter of the Sun, as to make it not appear larger than the Dog, or any other of the chief Stars. To this purpose I clos'd one End of my twelve-foot Tube with a very thin Plate, in the Middle of which I made a Hole not exceeding the twelfth Part of a Line, that is the hundred and forty fourth Part of an Inch. That End I turn'd to the Sun, placing my Eye at the other, and I could see so much of the Sun as was in Diameter about the 182d part of the Whole. But still that little piece of him was brighter much than the Dog-star is in the clearest Night. I saw that this would not do, but that must lessen the Diameter of the Sun a great deal more. I made [154]then such another Hole in a Plate, and against it I plac'd a little round Glass that I had made use of in my Microscopes, of much about the same Diameter with the former Hole. Then looking again towards the Sun (taking care that no Light might come near my Eye to hinder my Observation) I found it appeared of much the same Clearness with *Sirius*. But casting up my account, according to the Rules of *Dioptricks*, I found his Diameter now was but $\frac{1}{152}$ part of that hundred and eighty second part of his whole Diameter that I saw through the former Hole. Multiplying $\frac{1}{152}$ and $\frac{1}{182}$ into one another, the Product I found to be $\frac{1}{27664}$. The Sun therefore being contracted into such a Compass, or being removed so far from us (for it's the same thing) as to make his Diameter but the 27664 part of that we every Day see, will send us just the same Light as the Dog-star now doth. And his distance then from us will be to his present distance undoubtedly as 27664 is to 1; and his Diameter little above four Thirds, 4''.

Seeing [155]then *Sirius* is supposed equal to the Sun, it follows that his Diameter is likewise 4", and that his Distance to the Distance of the Sun from us is as 27664 to 1. And what an incredible Distance that is, will appear by the same way of reasoning that we used in measuring that of the Sun. For if 25 Years are required for a Bullet out of a Cannon, with its utmost Swiftneſs, to travel from the Sun to us; then by multiplying the Number 27664 into 25, we ſhall find that ſuch a Bullet would ſpend almoſt ſeven hundred thouſand Years in its Journey between us and the neareſt of the fix'd Stars. And yet when in a clear Night we look upon them, we cannot think them above ſome few Miles over our Heads. What I have here enquir'd into, is concerning the neareſt of them. And what a prodigious Number muſt there be beſides of thoſe which are placed in the vaſt Spaces of Heaven, as to be as remote from theſe as theſe are from the Sun! For if with our bare Eyes we can obſerve above a Thouſand, and with a Teleſcope can [156]diſcover ten or twenty times as many; what bounds of Number can we ſet to thoſe which are out of the Reach even of theſe Aſſiſtances! eſpecially if we conſider the infinite Power of God. Really, when I have been reflecting thus with my ſelf, my thoughts all our Arithmetick was nothing, and we are verſ'd but in the very Rudiments of Numbers, in compariſon of this great Sum. For this requires an immenſe Treafury, not of twenty or thirty Figures only, in our decuple Progreſſion, but of as many as there are Grains of Sand upon the Shore. And yet who can ſay, that even this Number exceeds that of the Fix'd Stars? Some of the Ancients, and *Jordanus Brunus* carry'd it further, in declaring the Number infinite: he would perſwade us that he has prov'd it by many Arguments, tho' in my opinion they are none of them concluſive. Not that I think the contrary can ever be made out. Indeed it ſeems to me certain, that the Univerſe is infinitely extended; but what God has been pleas'd to place beyond the Region of [157]the Stars, is as much above our Knowledge, as it is beyond our Habitation.

Or what if beyond ſuch a determinate Space he has left an infinite Vacuum; to ſhow, how inconfiderable all that he has made is, to what his Power could, had he ſo pleas'd, have produced? But I am falling, before I am aware, into that intricate Diſpute of Infinity: Therefore I ſhall wave this, and not, as ſoon as I am free of one, take upon me another difficult Task. All that I ſhall do more is to add ſomewhat of my Opinion concerning the whole World, as it is a Place for the Reception of the Suns or fix'd Stars, every one of which, I have ſhewed, may have their Planetary Syſtems about them.

Every Sun has a Vortex round it, very different from thoſe of Cartes. I am of Opinion then that every Sun is ſurrounded with a Whirl-pool or Vortex of Matter in a very ſwift Motion; tho' not in the leaſt like *Cartes's* either in their Bulk, or manner of Motion. For *Cartes* makes his ſo large, as every one of them to touch all the others round them, in a flat Surface, juſt as you have ſeen the Bladders that [158]Boys blow up in Soap-luſds do; and would have the

whole Vortex to move round the same way. But the Angles of every Vortex will be no small hindrance to such a Motion. Then the whole Matter moving round at once, upon the Axis as it were of a Cylinder, did not a little puzzle him in giving Reasons for the Roundness of the Sun: which however they may satisfy some People that do not consider them, really prove nothing of the Matter. In this æthereal Matter the Planets float, and are carried round by its Motion: and the thing that keeps them in their own Orbs is, that they themselves, and the Matter in which they swim, equally strive to fly off from the Center of this Motion. Against all which there are many Astronomical Objections, some of which I touch'd upon in my Essay of the Causes of Gravity. Where I gave another Account of the Planets not deserting their own Orbs; which is their Gravitation towards the Sun. I shew'd there the Causes of that Gravitation, and cannot but wonder that [159]*Cartes*, the first Man that ever began to talk reasonably of that Matter, should never meddle with, or light on it. *Plutarch* in his Book of the Moon above-mentioned says, that some of the Ancients were of Opinion, that the Reason of the Moon's keeping her Orbit was, that the Force of her Circular Motion was exactly equal to her Gravity, the one of which pull'd her to, as much as the other forc'd off from the Centre. And in our Age *Alphonfus Borellus*, who was of this same Opinion in the other Planets as well as the Moon, makes the Gravitation of the primary Planets to be towards the Sun, as that of the Secondary is towards the Planets round which they move: Which Sir *Isaac Newton* has more fully explain'd, with a great deal of Pains and Subtilty; and how from that Cause proceeds the Ellipticity of the Orbs of the Planets, found out by *Kepler*. According to my Notion of the Gravitation of the Planets to the Sun, the Matter of his Vortex must not at all move the same [160]way, but after such a manner as to have its Parts carry'd different ways on all Sides. And yet there is no fear of its being destroyed by such an irregular Motion, because the Æther round it, which is at rest, keeps the Parts of it from flying out. With the Help of such a Vortex as this I have undertook in that Essay to explain the Gravity of Bodies on this Earth, and all the Effects of it. And I suppose there may be the same Cause as well of the Gravitation of the Planets, and of our Earth among the rest, towards the Sun, as of their Roundness: A Thing so very hard to give an Account of in *Cartes's* System.

I must differ from him too in the Bigness of the Vortices, for I cannot allow them to be so large as he would make them. I would have them dispersed all about the immense Space, like so many little Whirl-pools of Water, that one makes by the stirring of a Stick in any large Pond or River, a great way distant from one another. And as their Motions do not all intermix [161]or communicate with one another, so in my Opinion must the Vortices of Stars be placed as not to hinder one another's free Circumrotations.

So that we may be secure, and never fear that they will swallow up or destroy one another; for that was a mere Fancy of *Cartes*'s, when he was shewing how a fix'd Star or Sun might be turn'd into a Planet. And 'tis plain that when he writ it, he had no Thoughts of the immense Distance of the Stars from one another; particularly, by this one Thing, that he would have a Comet as soon as ever it comes into our Vortex, to be seen by us. Which is as absurd as can be. For how could a Star, which gives us such a vast Light only from the Reflection of the Beams of the Sun, as he himself owns they do; how I say could that be so plainly seen at a distance Ten thousand times larger than the Diameter of the Earth's Orbit? He could not but know that all round the Sun there is a vast *Extensum*; so vast, that in *Copernicus*'s System the [162] *magnus Orbis* is counted but a Point in comparison with it. But indeed all the whole Story of Comets and Planets, and the Production of the World, is founded upon such poor and trifling Grounds, that I have often wonder'd how an ingenious Man could spend all that pains in making such Fancies hang together. For my part, I shall be very well contented, and shall count I have done a great Matter, if I can but come to any Knowledge of the Nature of Things, as they now are, never troubling my self about their Beginning, or how they were made, knowing that to be out of the reach of human Knowledge, or even Conjecture.

FINIS.

Colophon

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	The celestial worlds discovered: or, conjectures concerning the inhabitants, plants and productions of the worlds in the planets	
Title:		
Author:	Christiaan Huygens (1629– 1695)	Info https://viaf.org/viaf/9894043/
File generation date:	2023-07-14 20:24:52 UTC	
Language:	English	
Original publication date:	1722	

Revision History

2023-07-09 Started.

Corrections

The following corrections have been applied to the text:

Page	Source	Correction	Edit distance
11 , 11 , 112	[<i>Not in source</i>]	.	1
20	Gavitation	Gravitation	1
20	Carte's	Cartes's	1
25	argee	agree	2
37	prevaling	prevailing	1
42	there	they are	3
54	inflamable	inflammable	1
55	Defart	Defert	1
57	fefs'd	poffefs'd	3
74	cloath'd	cloth'd	1
76	necef-	neceffary	4
80	Pover-verty	Poverty	4
106	intollerably	intolerably	1
111	enlightned	enlighten'd	2
114	anothet	another	1
115	Unweildinefs	Unwieldinefs	2
116 , 116 , 116 , 116 , 116 , 116 , 116 , 116 , 116 , 116 , 124	[<i>Not in source</i>]	,	1
119	comes	come	1
127	Midnidght	Midnight	1
132	Is	is	1
147	anonother	another	2
151	encreafed	increafed	1

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